Shift-Symmetric Configurations in d-Dimensional Cellular Automata: Enumeration, Bounds, Irreversibility, and Detection

Peter Banda
Cequence

John Caughman
Portland State University

Tobi Lehman
Functorial, Inc.

Abstract

A configuration of a cellular automaton on the d-dimensional torus $\mathbb{Z}_n^d$ is *shift-symmetric* if it is fixed by some nonzero translation. Banda, Caughman, Cenek, and Teuscher (*Chaos* 29, 063120, 2019) enumerated these configurations in two dimensions and drew dynamical and algorithmic consequences. We extend their results to every dimension. We give an explicit closed-form enumeration for all n, d, and alphabet sizes σ, in which the two-dimensional coefficients 1, $p{+}1$, and p become $r_d(p,v) = p^{\binom{v}{2}} \begin{bmatrix} d \\ v \end{bmatrix}_p$; it extends to configurations with a prescribed number of active cells and to all finite abelian groups. The case of prime n recovers, in complement form, a result of Castillo-Ramirez and Gadouleau; the multi-prime closed form and the dynamical consequences beyond the stabilizer inclusion are new. We prove a lower bound attained *exactly* when n is prime, and upper bounds, among them $(2^d - 1 + 2\log_2 n)\,\sigma^{n^d/2}$. As d grows, the symmetry probability decays as $\sigma^{-c\,n^d}$ with $c \to 1 - 1/p_1(n)$, set solely by the smallest prime factor of n, and the symmetry advantage of the side $p{+}1$ over a prime side p inverts within an explicit window around $d_c(p) \approx p \ln 2$. Under the necessary hypothesis of translation-equivariant neighborhoods, shift-symmetry is irreversible, the per-prime symmetry ranks never decrease, and non-symmetric tasks are unsolvable from symmetric inputs. We give a detection algorithm with worst-case cost $O\big(n^d \sum_{p|n} \frac{p^d-1}{p-1}\big)$ and an average-case analysis, exact for each subgroup test. The mathematical content of every numbered statement is formally verified in Lean 4, and the correspondence between each theorem, lemma, proposition, and corollary and its formal statement was audited by hand.

Cellular automata are grids of cells that update in parallel, every cell applying the same local rule to its neighbors at the same relative positions. On a grid that wraps around like a torus, some patterns are unchanged when shifted by a nonzero offset. Such a uniform update can never destroy this symmetry, so tasks that require breaking it, such as electing a single leader cell, cannot be solved from these patterns. How likely is a random starting pattern to be one of them, and how does this depend on the grid? For two-dimensional grids, these questions were answered in 2019. Here we give an exact formula in every dimension. The arithmetic of the side length governs the answer: a simple formula bounds the count from below and is exact precisely when the side length is prime, and as the dimension grows, the smallest prime factor of the side length alone sets the rate per cell at which such patterns become rare. All theorems were also verified by computer in the Lean proof assistant.

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1 Introduction

A configuration of a cellular automaton on the d-dimensional torus $\mathbb{Z}_n^d$ is said to be *shift-symmetric* if it is fixed by some nonzero translation (Figure 1). For the two-dimensional case, Banda, Caughman, Cenek, and Teuscher (*Chaos* 29, 063120, 2019) studied shift-symmetry for configurations of two-dimensional cellular automata (CA) on toroidal lattices $\mathbb{Z}_n \times \mathbb{Z}_n$. Such symmetry is noteworthy since, once a configuration is fixed by a nonzero translation, no uniform synchronous update can ever cause the symmetry to be lost. As a result, symmetric configurations effectively function as irreversible traps that can make non-symmetric tasks (leader election, pattern recognition) unsolvable in the worst case. Quantifying this obstruction requires counting the shift-symmetric configurations, which Banda et al. [1] accomplished for $d = 2$, leaving higher-dimensional extensions for future consideration. In this paper, we provide the d-dimensional extension and, in the process, sharpen and correct several statements whose naive generalizations to higher dimensions fail.

Contributions. The primary contributions of this paper are the following:

(i) An explicit closed-form enumeration of the set S_{n^d} of shift-symmetric configurations on $\mathbb{Z}_n^d$ for arbitrary n, d, and alphabet size σ (Theorem 4.4), with two independent derivations, and a refinement to configurations with exactly k active cells (Theorem 5.1); both extend to every finite abelian group, in particular to rectangular tori, with d replaced by the p-ranks (Theorem 4.7, Remark B.2), and the count is smallest for the cyclic group among the abelian groups of a given order (Corollary B.1).

(ii) A lower bound for $|S_{n^d}|$ attained exactly when n is prime (Theorem 6.1). We show that a naive generalization of the two-dimensional bound of [1] is *false* for $d \geq 3$ (Remark 6.2). We also give an upper bound with an explicit prefactor $2^d - 1 + 2\log_2 n$ (Theorem 6.3).

(iii) Proofs of irreversibility, insolvability, and per-prime symmetry-rank monotonicity in d dimensions (Section 8), under a translation-equivariance hypothesis that we show is *necessary*: a two-cell counterexample (Remark 2.5) demonstrates that a shared local rule with cell-dependent neighborhoods does not preserve symmetry.

(iv) A detection algorithm for shift-symmetry with worst-case bound $O(\mathrm{sopf}_d(n) \cdot n^d)$ and an exact average-case expectation analysis (Section 9), with the average-case "polylogarithmic" claims of the two-dimensional literature restated under appropriate hypotheses.

(v) A number of computational observations regarding higher dimensions (Section 10): the prime/composite extremal pattern of $d = 2$ *breaks* for $d \geq 3$, and the multi-term correction regime collapses rapidly with growing p and d.

Comments on verification. Every enumerative formula below has been checked, with exact integer arithmetic, against definition-level exhaustive enumeration and against an independent subgroup-level inclusion–exclusion evaluator; the two main theorems additionally passed independent adversarial review. The supplementary material, namely the scripts that reproduce every enumeration, scan, and figure (with their domains, outputs, and checksums),[1] the Lean development (Section 11), and the correspondence audit (Section 12), is in the project repository, `github.com/peterbanda/d-dim-shift-symmetry-in-ca`. Beyond testing and review, the numbered results are *machine-checked*: Theorems 4.4 and 6.1, in the generality stated here, and the mathematical content of the other numbered statements of Sections 2–10 and Appendices A

[1] Catalogued in `research/experiments/`; see the reproducibility paragraph in Section 10. A companion erratum to [1], correcting misprints and gaps in its proofs, is in `research/2d_fix_note/`.

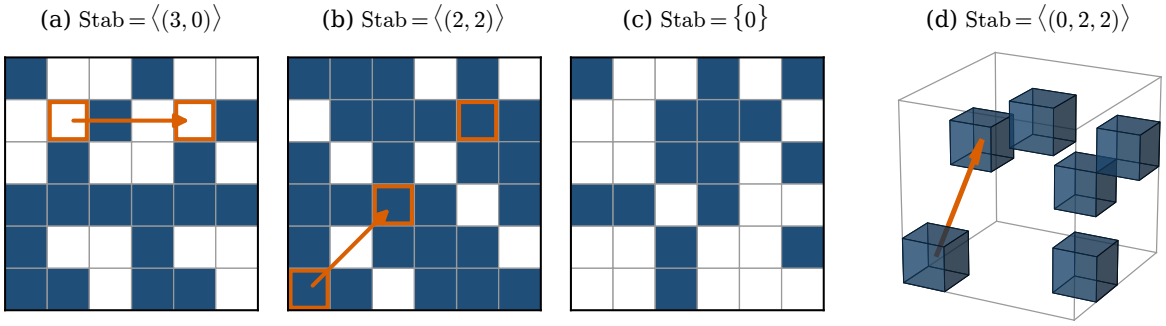


Figure 1: Shift-symmetric and asymmetric configurations ($\sigma = 2$: filled and empty cells). A cell $\mathbf{u} = (u_1, u_2)$ is drawn in column u_1 and row u_2, counted from the bottom left, and $\mathbf{s}$ is fixed by $\mathbf{v}$ when $\mathbf{s}(\mathbf{u}+\mathbf{v}) = \mathbf{s}(\mathbf{u})$ for all $\mathbf{u}$; each arrow runs from a cell $\mathbf{u}$ to $\mathbf{u}+\mathbf{v}$ for a generator $\mathbf{v}$ of the stabilizer. (a) A configuration on $\mathbb{Z}_6^2$ with stabilizer $\langle(3,0)\rangle$: the translation by $(3,0)$ maps every cell to a cell in the same state (outlined: one orbit). (b) Stabilizer $\langle(2,2)\rangle$, a diagonal translation of order 3. (c) Trivial stabilizer: this configuration is not shift-symmetric. (d) A configuration on $\mathbb{Z}_4^3$, its six filled cells drawn, with stabilizer $\langle(0,2,2)\rangle$; the arrow runs from $(0,0,0)$ to $(0,2,2)$. Of the 2^{36} configurations on $\mathbb{Z}_6^2$, exactly 801 064 (about $1.2 \cdot 10^{-5}$ of them) are shift-symmetric (Theorem 4.4; a reference value of Section 4.4).

and B (statistical summaries aside) have been formally verified in the Lean 4 proof assistant (Section 11).

Related work. For the two-dimensional torus, Banda et al. [1] enumerated shift-symmetric configurations, bounded their number, and analyzed detection; to our knowledge no extension has appeared since. Earlier, CA leader election had been studied in one dimension, both by evolving binary rules [29] and through symmetry-based impossibility results and performance upper bounds [30]. From the group-theoretic side, Castillo-Ramirez and Gadouleau [2] counted configurations with a given stabilizer class over an *arbitrary* finite group via the Möbius function of the subgroup lattice. Their Lemma 7, which uses Hall's evaluation of the Möbius function of a p-group ([7]; see Hawkes, Isaacs, and Özaydin [6, Cor. 3.5]; see also Delsarte [13] for Möbius inversion over the subgroups of finite abelian groups), yields in complement form the *prime-power* case $n = p^\alpha$ of our enumeration theorem: by its part (iii), it remains to count the elementary abelian subgroups of $\mathbb{Z}_{p^\alpha}^d$, which are the subspaces of its p-torsion $\cong \mathbb{F}_p^d$ (Lemma 3.3); part (iv) is the prime case. Section 4.3 shows how the general case can alternatively be derived from their framework. What appears neither there nor in the least-period counts of Castillo-Ramirez and Sánchez-Álvarez [4], whose worked examples stop at rank two, is the explicit multi-prime closed form for $\mathbb{Z}_n^d$ and for finite abelian groups (Section 4.5), the prime-tightness characterization for $d \geq 2$ (for $d = 1$ it is noted after [2, Lemma 7]), or the consequences drawn in Section 8 (insolvability, rank monotonicity). On the combinatorial side, Ethier [9] and Ethier and Lee [10] count two-dimensional toroidal arrays up to symmetry (orbits, not symmetric configurations), and Adamson, Deligkas, Gusev, and Potapov study multidimensional necklaces algorithmically and state a recursion across dimensions for the number of "atranslational" ones — the orbit-level complement of our shift-symmetric configurations — on rectangular grids [8, Thm. 3];[2] they derive no non-recursive formula for this count (it is $(\sigma^N - |S_G|)/N$ with $|S_G|$ from Theorem 4.7; see Remark B.2; see also the group-theoretic necklace polynomials of Oh [14]) and no dynamical consequences. In symbolic dynamics, the numbers of points fixed by a *given* finite-index subgroup $L \leq \mathbb{Z}^d$ (for the full shift, $\sigma^{[\mathbb{Z}^d:L]}$) are classical and are packaged

[2] As printed, this recursion does not reproduce exhaustive counts: none of the readings of its formula that we tried matches them on all grids with at most 16 cells over two and three symbols (Run 037 of the repository).

by Lind's zeta function for $\mathbb{Z}^d$-actions [11]; Miles and Ward [12] invert them over the lattice of finite-index subgroups to count the orbits with a given stabilizer, and study the growth of these counts, without reducing them to the explicit rank-indexed formula used here. Our object is, instead, the union over all nontrivial translations on a fixed finite torus. Lower bounds on the number of aperiodic configurations of a general finite group were given by Gao, Jackson, and Seward [5, Cor. 1.7.2] and sharpened in [2, Thm. 5]. On $\mathbb{Z}_n^d$, the former is exponentially weaker in the exponent than Proposition 10.1; the latter, whose proof contains our union bound (Theorem 6.3(i)), already implies its limit, with a larger prefactor for $n \geq 7$. For $d = 1$ our formulas reduce to classical aperiodic-word counts going back to Moreau and Witt [24, 25]; see also [23]. The prime-lattice, binary, two-dimensional case of our count also appears, without proof, as a formula line in OEIS entry A323863 [15] (orbit-level companion entries: A323872, A184271). On the dynamical side, Kornyak [16] observed that deterministic discrete dynamics cannot decrease symmetry, and for cellular automata on a finite group the inclusion of the stabilizer of a configuration in that of its image (Theorem 8.1) is stated in the proof of [2, Lemma 10], with equality for invertible ones in that of [2, Lemma 3]; its converse (some cellular automaton maps x to y if $\mathrm{Stab}(x) \leq \mathrm{Stab}(y)$) is [2, Lemma 10] for distinct orbits and [3, Thm. 2(iii)] in general. The orbit count of Remark B.2(ii) is the case $H = \{e\}$ of [2, Lemma 4].

2 Model

In this section we introduce some of the basic definitions and notation we will use in our study of cellular automata and their configurations.

Definition 2.1 (Lattice and configurations)**.** *Fix integers $n \geq 2$, $d \geq 1$ and a finite alphabet Σ with $\sigma = |\Sigma| \geq 2$ (for counting statements, $\sigma = 1$ is also allowed). We define the* **lattice** $\mathcal{A}$ *to be the finite abelian group $\mathcal{A} = (\mathbb{Z}/n\mathbb{Z})^d$, and we refer to the $N = n^d$ elements of $\mathcal{A}$ as the* **cells** *of $\mathcal{A}$. A* **configuration** *is any function $\mathbf{s} : \mathcal{A} \to \Sigma$.*

Definition 2.2 (Translations acting on configurations)**.** *For $\mathbf{v} \in \mathcal{A}$, the* **translation** $\tau_{\mathbf{v}}$ *acts on the set of configurations by $(\tau_{\mathbf{v}}\mathbf{s})(\mathbf{u}) = \mathbf{s}(\mathbf{u} + \mathbf{v})$, for any $\mathbf{u}, \mathbf{v} \in \mathcal{A}$. For any configuration $\mathbf{s}$, the corresponding* **stabilizer***, denoted $\mathrm{Stab}(\mathbf{s}) = \{\mathbf{v} \in \mathcal{A} : \tau_{\mathbf{v}}\mathbf{s} = \mathbf{s}\}$, is a subgroup of $\mathcal{A}$. For any subgroup $H \leq \mathcal{A}$, its* **fixed set** *is given by $\mathrm{Fix}(H) = \{\mathbf{s} : \tau_{\mathbf{h}}\mathbf{s} = \mathbf{s}$ for all $\mathbf{h} \in H\}$.*

Definition 2.3 (Shift-symmetry)**.** *For $\mathbf{0} \neq \mathbf{v} \in \mathcal{A}$, we define*

$$S_{n^d}(\mathbf{v}) = \{\mathbf{s} : \tau_{\mathbf{v}}\mathbf{s} = \mathbf{s}\} = \mathrm{Fix}(\langle \mathbf{v} \rangle).$$

The set of **shift-symmetric configurations** *is*

$$S_{n^d} \;=\; \bigcup_{\mathbf{0} \neq \mathbf{v} \in \mathcal{A}} S_{n^d}(\mathbf{v}) \;=\; \{\mathbf{s} : \mathrm{Stab}(\mathbf{s}) \neq \{\mathbf{0}\}\}.$$

Definition 2.4 (Equivariant cellular automaton)**.** *A* **cellular automaton** *on $\mathcal{A}$ over Σ is given by a finite list of* **offsets** $\mathbf{o}_1, \ldots, \mathbf{o}_b \in \mathcal{A}$ *and a* **local rule** $\phi : \Sigma^b \to \Sigma$*; the* **global map** $\Phi : \Sigma^{\mathcal{A}} \to \Sigma^{\mathcal{A}}$ *updates all cells synchronously:*

$$\Phi(\mathbf{s})(\mathbf{u}) \;=\; \phi\big(\mathbf{s}(\mathbf{u} + \mathbf{o}_1), \ldots, \mathbf{s}(\mathbf{u} + \mathbf{o}_b)\big).$$

Equivalently, Φ is **translation-equivariant***: $\Phi(\tau_{\mathbf{v}}\mathbf{s}) = \tau_{\mathbf{v}}\Phi(\mathbf{s})$ for all $\mathbf{v} \in \mathcal{A}$.*

Remark 2.5 (Equivariance is necessary, not cosmetic). We note that the property of translation-equivariance is used extensively in our discussion of dynamical results. Indeed, definitions in the style of "a neighborhood function $\eta : \mathcal{A} \times \Sigma^{\mathcal{A}} \to \Sigma^b$ together with a shared local rule ϕ" do *not* imply equivariance and do *not* support the irreversibility theorem, because η may depend on the cell arbitrarily.

Here is a counterexample for state-valued η ($d = 1$, $n = 2$, $\Sigma = \{0, 1\}$, $b = 1$): let $\eta_0(\mathbf{s}) = (s_0)$, $\eta_1(\mathbf{s}) = (0)$, and $\phi = \mathrm{id}$. All cells share ϕ, yet the shift-symmetric configuration $(1, 1)$ maps to $(1, 0)$, which is not shift-symmetric.

If one insists that η only select *locations* (each cell reads actual cells, just not translation-uniform ones), equivariance can still fail on every $\mathcal{A}$, and some location-selecting map breaks irreversibility exactly when $|\mathcal{A}| = n^d$ is not prime (on $\mathbb{Z}_p$, p prime, every shift-symmetric configuration is constant and is mapped to a constant one). In $d = 1$ the smallest case is $n = 4$: with $\phi = \mathrm{id}$, let every cell read itself except that cell 2 reads cell 1; then $(0, 1, 0, 1)$, fixed by the shift 2, maps to $(0, 1, 1, 1)$, which is not.

The original paper by Banda et al. [1] considers the two-dimensional case and avoids this pitfall only inside its proofs, where the neighborhood is explicitly assumed to be given by shared relative offsets [1, Lemma A.5]. Our Definition 2.4 in this paper makes the hypothesis explicit from the start.

3 Preliminaries

In this section, we review some preliminary results about how groups of translations act on the set of configurations.

Throughout, we refer to the prime factorization $n = \prod_{i=1}^{k} p_i^{\alpha_i}$, with $p_1 < \cdots < p_k$ prime and integers $\alpha_1, \ldots, \alpha_k \geq 1$. We note that the number of distinct prime factors of n is $k = \omega(n)$. Also recall that, for any prime p, the Gaussian binomial $\begin{bmatrix} d \\ r \end{bmatrix}_p = \prod_{i=0}^{r-1} \frac{p^{d-i}-1}{p^{i+1}-1}$ gives the number of r-dimensional subspaces of $\mathbb{F}_p^d$, which is the d-dimensional vector space over $\mathbb{F}_p$, the finite field with p elements.

Lemma 3.1 (Fixed sets and generated subgroups)**.** *For any given subgroup $H \leq \mathcal{A}$ and any given finite family of subgroups $H_1, \ldots, H_\ell \leq \mathcal{A}$:*

(i) $|\operatorname{Fix}(H)| = \sigma^{|\mathcal{A}|/|H|}$;

(ii) $\bigcap_{a=1}^{\ell} \operatorname{Fix}(H_a) = \operatorname{Fix}\big(\langle H_1 \cup \cdots \cup H_\ell \rangle\big)$.

Proof. (i) A configuration $\mathbf{s} \in \operatorname{Fix}(H)$ iff $\mathbf{s}$ is constant on the cosets of H. Translation by H on $\mathcal{A}$ is a free action, so its orbits are the $|\mathcal{A}|/|H|$ cosets, giving a bijection $\operatorname{Fix}(H) \cong \Sigma^{\mathcal{A}/H}$.

(ii) A configuration $\mathbf{s}$ lies in the left side iff $H_1 \cup \cdots \cup H_\ell \subseteq \operatorname{Stab}(\mathbf{s})$. This happens iff the generated subgroup is contained in $\operatorname{Stab}(\mathbf{s})$, since $\operatorname{Stab}(\mathbf{s})$ is a subgroup. □

Lemma 3.2 (Order of a translation)**.** *For $\mathbf{0} \neq \mathbf{v} = (l_1, \ldots, l_d) \in \mathcal{A}$, the order of $\mathbf{v}$ as an element of the additive group $\mathcal{A}$ is given by*

$$|\langle \mathbf{v} \rangle| = \operatorname{lcm}_i\big(n/\gcd(l_i, n)\big) = n/\gcd(l_1, \ldots, l_d, n),$$

and

$$|S_{n^d}(\mathbf{v})| = \sigma^{\,n^d/|\langle \mathbf{v} \rangle|} = \sigma^{\,n^{d-1}\gcd(l_1,\ldots,l_d,n)}.$$

Proof. The order of $\mathbf{v}$ is the least integer $t \geq 1$ with $t \cdot l_i \equiv 0 \pmod{n}$ for all i. So $|\langle \mathbf{v} \rangle|$ must equal $\operatorname{lcm}_i(n/\gcd(l_i, n))$. Comparing p-adic valuations, $v_p(\operatorname{lcm}_i n/\gcd(l_i, n)) = v_p(n) - \min_i \min(v_p(l_i), v_p(n)) = v_p(n/\gcd(l_1, \ldots, l_d, n))$ for every prime $p \mid n$. The count is the case $H = \langle \mathbf{v} \rangle$ of Lemma 3.1(i) above. □

Lemma 3.3 (Torsion and lines)**.** *Let $p \mid n$ be prime. The following hold.*

(i) The p-torsion $\mathcal{A}[p] = \{\mathbf{x} : p\mathbf{x} = \mathbf{0}\}$ is given by

$$\mathcal{A}[p] = \big(\tfrac{n}{p}\mathbb{Z}/n\mathbb{Z}\big)^d.$$

(ii) The map $\mathbf{x} = \frac{n}{p}\,\mathbf{y} \mapsto \mathbf{y} \bmod p$ *is an isomorphism* $\mathcal{A}[p] \cong \mathbb{F}_p^d$ *of* $\mathbb{F}_p$*-vector spaces.*

(iii) Under this isomorphism,

- *elements of order* p *correspond to nonzero vectors;*
- *cyclic subgroups of order* p *correspond to lines, and their number is* $\begin{bmatrix} d \\ 1 \end{bmatrix}_p = (p^d-1)/(p-1)$*;*
- *for a set* J *of order-*p *cyclic subgroups,*

$$|\langle \bigcup J \rangle| = p^{\operatorname{rk} J},$$

where $\operatorname{rk} J$ *is the dimension of the span of the corresponding lines.*

Proof. Componentwise, $px = 0$ in $\mathbb{Z}/n\mathbb{Z}$ iff $x \in \frac{n}{p}\mathbb{Z}/n\mathbb{Z}$. The group $\mathcal{A}[p]$ is an elementary abelian p-group, and therefore it is an $\mathbb{F}_p$-vector space. The subgroups of $\mathcal{A}[p]$ are its subspaces, and a subgroup generated by a union of lines is their span. □

Lemma 3.4 (Completeness)**.** *For any prime* p*, let* $\mathcal{L}_p$ *be the set of cyclic subgroups of order* p *of* $\mathcal{A}$ *and let* $\mathcal{L} = \bigcup_{p|n} \mathcal{L}_p$*. Then*

$$S_{n^d} = \bigcup_{C \in \mathcal{L}} \operatorname{Fix}(C).$$

Proof. Since S_{n^d} contains $\operatorname{Fix}(H)$ for *all* nontrivial cyclic subgroups H of $\mathcal{A}$, the direction $\supseteq$ is clear. For the other direction $\subseteq$, we argue as follows. If a configuration $\mathbf{s}$ has a nonzero $\mathbf{w} \in \operatorname{Stab}(\mathbf{s})$, let $o > 1$ be its order. Then o divides the exponent n of $\mathcal{A}$, so any prime divisor p of o must also divide n, and $(o/p)\mathbf{w}$ has order exactly p. It follows that the cyclic subgroup $C := \langle (o/p)\mathbf{w} \rangle \in \mathcal{L}_p$ satisfies $C \leq \operatorname{Stab}(\mathbf{s})$, and so $\mathbf{s} \in \operatorname{Fix}(C)$. □

Lemma 3.5 (Multi-prime generated subgroups)**.** *Let* $J = \bigsqcup_{p|n} J_p$ *with* $J_p \subseteq \mathcal{L}_p$*. Then*

$$|\langle \bigcup J \rangle| = \prod_{p|n} p^{\operatorname{rk} J_p}.$$

Proof. Write $\mathcal{A} = \bigoplus_{p|n} \mathcal{A}_p$ (primary decomposition). Each $C \in \mathcal{L}_p$ lies in $\mathcal{A}[p] \subseteq \mathcal{A}_p$, so $\langle \bigcup J \rangle = \bigoplus_p \langle \bigcup J_p \rangle$, and each factor has order $p^{\operatorname{rk} J_p}$ by Lemma 3.3. □

4 Enumeration

In this section, we apply the results above to count the number of shift-symmetric configurations in S_{n^d} for arbitrary finite dimension d.

4.1 The signed spanning-set count

We begin by introducing a function to denote an alternating sum that will help us to study the various sets of lines that span a finite vector space.

Lemma 4.1. *Let* p *be prime,* $d \geq 1$*, and* $V \leq \mathbb{F}_p^d$ *a subspace of dimension* r*, with* $\operatorname{lines}(V)$ *its set of* 1*-dimensional subspaces. Define*

$$f(V) \;:=\; \sum_{\substack{J \subseteq \operatorname{lines}(V) \\ \operatorname{span} J = V}} (-1)^{|J|}$$

Then

$$f(V) \;=\; \mu(0, V) \;=\; (-1)^r p^{\binom{r}{2}},$$

where μ *is the Möbius function of the subspace lattice [26, Ex. 3.10.2].*

Proof. Every $J \subseteq \text{lines}(V)$ spans exactly one $W \leq V$, so

$$\sum_{W \leq V} f(W) = \sum_{J \subseteq \text{lines}(V)} (-1)^{|J|} = (1-1)^{|\text{lines}(V)|} = \delta_{V,0},$$

since $\text{lines}(V) \neq \emptyset$ exactly when $V \neq 0$. By Möbius inversion on the finite subspace lattice, we have $f(V) = \mu(0, V)$. (For more on inversion, see [26, Prop. 3.7.1] and [27, §4.1.2]).

Alternatively, an argument using the q-binomial theorem [26, Eq. (1.87)] can be given as follows. The function mapping a vector space V to $(-1)^{\dim V} p^{\binom{\dim V}{2}}$ satisfies the same recursion as f above, because, by the q-binomial theorem at $x = -1$, we have

$$\sum_{r=0}^{R} \begin{bmatrix} R \\ r \end{bmatrix}_p (-1)^r p^{\binom{r}{2}} = \prod_{i=0}^{R-1} (1 - p^i) = \delta_{R,0}.$$

This recursion determines f uniquely by induction on $\dim V$. □

Using the lemma above, we can readily evaluate a similar sum that is taken over all of the sets of lines in $\mathbb{F}_p^d$ whose span has a given rank r.

Lemma 4.2. *Let p be prime, $d \geq 1$, and define*

$$c_r(p) \;:=\; \sum_{\substack{J \subseteq \text{lines}(\mathbb{F}_p^d) \\ \text{rk}\, J = r}} (-1)^{|J|}$$

Then

$$c_r(p) \;=\; (-1)^r p^{\binom{r}{2}} \begin{bmatrix} d \\ r \end{bmatrix}_p, \qquad 0 \leq r \leq d. \tag{1}$$

Proof. Grouping the sets J with $\text{rk}\, J = r$ by their span, an r-dimensional subspace W, and summing $f(W) = (-1)^r p^{\binom{r}{2}}$ (Lemma 4.1) over the $\begin{bmatrix} d \\ r \end{bmatrix}_p$ subspaces of dimension r gives (1). □

4.2 Main theorem

We are now ready to prove our main result. We begin by defining the coefficient function that appears in our expression for the number of shift-symmetric configurations in S_{n^d} for arbitrary finite dimension d.

Definition 4.3 (Coefficient function)**.** *For a prime p and integers $0 \leq v \leq d$, let*

$$r_d(p, v) = p^{\binom{v}{2}} \begin{bmatrix} d \\ v \end{bmatrix}_p,$$

with $r_d(p, 0) = 1$.

Remark. In the case when $d = 2$, we observe that $r_2(p, v) = 1, p+1, p$ for $v = 0, 1, 2$, which matches the coefficients given in [1, Thm. E.19].

Theorem 4.4 (Enumeration in d dimensions)**.** *For all $n \geq 2$, $d \geq 1$, and $\sigma \geq 1$:*

$$\boxed{|S_{n^d}| \;=\; \sum_{\substack{\mathbf{v}=(v_1,\ldots,v_k) \in \{0,\ldots,d\}^k \\ \mathbf{v} \neq \mathbf{0}}} (-1)^{1+|\mathbf{v}|} \prod_{i=1}^{k} r_d(p_i, v_i)\; \sigma^{\,n^d \prod_i p_i^{-v_i}}} \tag{2}$$

where $|\mathbf{v}| = \sum_i v_i$. All exponents are integers.

Proof. By Lemma 3.4 and inclusion–exclusion, using Lemma 3.1 and then Lemma 3.5,

$$|S_{n^d}| \;=\; \sum_{\emptyset\neq J\subseteq\mathcal{L}} (-1)^{|J|+1}\big|\mathrm{Fix}(\langle\bigcup J\rangle)\big| \;=\; \textstyle\sum_{\emptyset\neq J}(-1)^{|J|+1}\,\sigma^{\,N/\prod_i p_i^{\,\mathrm{rk}\,J_{p_i}}}.$$

The exponent of σ depends only on the rank vector $\mathbf{v}(J) = (\mathrm{rk}\,J_{p_1},\dots,\mathrm{rk}\,J_{p_k})$; the sign is handled by the factorization below. Adding and subtracting a term σ^N for the empty set, we obtain

$$|S_{n^d}| \;=\; -\Big[\sum_{\text{all } J}(-1)^{|J|}\,\sigma^{N/\prod p_i^{v_i(J)}} \;-\; \sigma^N\Big].$$

Since $(-1)^{|J|} = \prod_i (-1)^{|J_{p_i}|}$ and the parts J_{p_i} vary independently, the sum factors over rank vectors, with per-prime factors $c_{v_i}(p_i)$ computed inside $\mathbb{F}_{p_i}^d$ via Lemma 3.3 and Lemma 4.2 (note $c_0 = 1$):

$$\sum_{\text{all } J}(-1)^{|J|}\,\sigma^{N/\prod p_i^{v_i}} \;=\; \sum_{\mathbf{v}\in\{0,\dots,d\}^k}\;\prod_{i=1}^{k} c_{v_i}(p_i)\;\sigma^{N/\prod_i p_i^{v_i}}.$$

The $\mathbf{v}=\mathbf{0}$ term is σ^N and cancels, and $-\prod_i c_{v_i}(p_i) = (-1)^{1+|\mathbf{v}|}\prod_i r_d(p_i,v_i)$, giving (2). As for the integrality of the exponents, note that $v_i \le d$ and $p_i \mid n$ imply $\prod_i p_i^{v_i} \mid n^d$. □

Corollary 4.5 (Prime lattices)**.** *Fix any prime $n = p$ and integers $d \ge 1$ and $\sigma \ge 1$. Then*

$$|S_{p^d}| = \sum_{r=1}^{d}(-1)^{r+1}\,p^{\binom{r}{2}}\begin{bmatrix} d \\ r \end{bmatrix}_p \sigma^{\,p^{d-r}}.$$

In particular,

$$|S_p| = \sigma \quad (d=1),$$

$$|S_{p^2}| = (p{+}1)\sigma^p - p\sigma \quad (d=2),\ \text{and}$$

$$|S_{p^3}| = (p^2{+}p{+}1)\sigma^{p^2} - p(p^2{+}p{+}1)\sigma^p + p^3\sigma \quad (d=3).$$

Remark 4.6. We observe the following attributions and boundary checks.

(i) In complement form ($\mathrm{ac} = \sigma^N - |S|$), Corollary 4.5 is Lemma 7(iv) of Castillo-Ramirez and Gadouleau [2]; our multi-prime formula (2) of Theorem 4.4 does not appear there, but we will demonstrate how it can be derived from their framework in Section 4.3 below.

(ii) In a boundary case where $n = 1$, both sides vanish. In a boundary case where $\sigma = 1$, the per-prime identity $\sum_{r=0}^{d} c_r(p) = 0$ correctly makes the right side equal to 1.

(iii) The case for $d = 1$ recovers the classical count of non-primitive words (proper powers u^j, $j \ge 2$; equivalently, words, i.e. configurations on $\mathbb{Z}_n$, fixed by a nontrivial cyclic rotation). This is the complement $\sigma^n - \sum_{m|n}\mu(m)\,\sigma^{n/m}$ (μ the number-theoretic Möbius function) of the number of primitive words; the latter number is n times the Moreau–Witt count of primitive necklaces [24, 25].

(iv) The case for $d = 2$ recovers [1, Thm. E.19] exactly; for prime n it is [1, Cor. E.20].

(v) For $d = 2$, $\sigma = 2$, prime n, the complement of Corollary 4.5 appears as an unproved formula line in OEIS A323863 [15].

4.3 Alternative derivation via the subgroup lattice

Now we describe how an enumeration matching the above result can also be derived using an alternative line of reasoning.

In [2, Lemma 7(i)], Castillo-Ramirez and Gadouleau show that, for any finite group G of order $N \ge 2$ (for $N = 1$ the formula is trivial) and $|A| = q \ge 2$, the number of aperiodic configurations is $\mathrm{ac}(G;A) = \sum_{K\le G}\mu(\{e\},K)\,q^{N/|K|}$, where μ denotes the Möbius function of

the subgroup lattice. For $G = \mathcal{A}$, every subgroup splits along the primary decomposition ($K = \bigoplus_p (K \cap \mathcal{A}_p)$: the primary components of any $\mathbf{x} \in K$ are integer multiples of $\mathbf{x}$ by CRT, hence lie in K), the subgroup lattice is the direct product of the primary lattices, and μ is multiplicative across it [26, Prop. 3.8.2]. On each p-group factor $\mathcal{A}_p \cong (\mathbb{Z}/p^\alpha\mathbb{Z})^d$, by the theorem of Hall [7] (see [6, Cor. 3.5]), $\mu(\{e\}, K_p)$ vanishes unless K_p is elementary abelian — hence contained in the p-torsion $\cong \mathbb{F}_p^d$ — in which case $\mu(\{e\}, K_p) = (-1)^r p^{\binom{r}{2}}$ for rank r, and there are $\begin{bmatrix} d \\ r \end{bmatrix}_p$ such subgroups. Summing $|S_{n^d}| = \sigma^N - \mathrm{ac}(\mathcal{A}; \Sigma)$ by rank vectors reproduces (2). This second derivation is logically independent of Lemmas 3.4–4.2.

4.4 Computational verification

Formula (2) was verified by exact integer computation against (L1) definition-level brute force over all σ^{n^d} configurations and all nonzero shifts, on 29 parameter triples (all with $n^d \le 16$), and (L2) an independent inclusion–exclusion over all subsets of brute-force-discovered prime-order subgroups with BFS subgroup closures, on these and 15 further triples (44 in all), covering $d \le 4$, $n \le 60$ (including the multi-prime $n = 6, 10, 12, 15, 30, 60$ and prime powers $4, 8, 9$), and $\sigma \le 4$; reference values ($\sigma = 2$) include

$$|S_{2^2}| = 8, \quad |S_{3^2}| = 26, \quad |S_{5^2}| = 182, \quad |S_{2^3}| = 72, \quad |S_{3^3}| = 6398, \quad |S_{6^2}| = 801064.$$

Table 1: Coefficients $r_d(p, v) = p^{\binom{v}{2}} \begin{bmatrix} d \\ v \end{bmatrix}_p$ for small d.

d	$v=0$	$v=1$	$v=2$	$v=3$	$v=4$
1	1	1			
2	1	$p+1$	p		
3	1	p^2+p+1	$p(p^2+p+1)$	p^3	
4	1	p^3+p^2+p+1	$p(p^2+1)(p^2+p+1)$	$p^3(p^3+p^2+p+1)$	p^6

4.5 Finite abelian groups

The torus $\mathbb{Z}_n^d$ enters the proof of Theorem 4.4 only through its order and its p-torsion, which is $\mathbb{F}_p^d$ for every prime $p \mid n$ (Lemma 3.3). For a finite abelian group G of order N, written additively, let $G[p] = \{g \in G : pg = 0\}$, an $\mathbb{F}_p$-vector space, and let $d_p = \dim_{\mathbb{F}_p} G[p]$ be the *p-rank* of G. Configurations $G \to \Sigma$, translations, stabilizers, and S_G are defined as in Section 2 with $\mathcal{A}$ replaced by G; a rectangular torus $\mathbb{Z}_{n_1} \times \cdots \times \mathbb{Z}_{n_m}$ has $N = \prod_i n_i$ and $d_p = |\{i : p \mid n_i\}|$.

Theorem 4.7 (Enumeration on a finite abelian group)**.** *For every finite abelian group G of order N and every $\sigma \ge 1$,*

$$|S_G| \;=\; \sum_{\substack{\mathbf{v}=(v_p)_{p|N},\ 0 \le v_p \le d_p \\ \mathbf{v} \neq \mathbf{0}}} (-1)^{1+|\mathbf{v}|} \prod_{p|N} r_{d_p}(p, v_p)\, \sigma^{N/\prod_p p^{v_p}}.$$

For $G = \mathbb{Z}_n^d$, where $d_p = d$ for every $p \mid n$, this is Theorem 4.4.

For an abelian p-group, Theorem 4.7 is [2, Lemma 7(iii)] in complement form, the elementary abelian subgroups being the subspaces of $G[p]$.

Proof. The proof of Theorem 4.4 applies with $\mathcal{A}$ replaced by G and $\mathbb{F}_p^d$ by $G[p] \cong \mathbb{F}_p^{d_p}$: Lemmas 3.1 and 3.4 use only that G is a finite abelian group; a subgroup generated by subgroups of prime order lies in the socle $\bigoplus_{p|N} G[p]$ and is the direct sum of its intersections with the $G[p]$, which gives Lemma 3.5; and Lemmas 4.1–4.2 apply to each $G[p]$ with d replaced by d_p. For a rectangular torus, $G[p] = \prod_i \mathbb{Z}_{n_i}[p]$, and $\mathbb{Z}_n[p] \cong \mathbb{Z}_p$ if $p \mid n$, while $\mathbb{Z}_n[p] = 0$ otherwise. □

By Theorem 4.7, $|S_G|$ depends on G only through N and its p-ranks; for $\sigma \geq 2$, at fixed N it strictly increases with each p-rank, so the N cells arranged as a one-dimensional ring $\mathbb{Z}_N$ carry no more shift-symmetric configurations than any torus with N cells (Corollary B.1 in Appendix B, where Remark B.2 also extends Theorem 5.1 to G and counts the G-orbits of aperiodic configurations).

5 Configurations with k active cells

In this section we describe how to refine the enumeration of Theorem 4.4 to account for the number of active cells.

To this end, fix any state $a \in \Sigma$ and let $S^a_{n^d,k}$ be the set of shift-symmetric configurations with exactly k cells in state a. In Theorem 5.1, its proof, and Corollary 7.2 (with $h = h(\mathbf{v})$ in Theorem 5.1, $h = |H|$ in its proof, and $h = 1$ in Corollary 7.2), $\binom{N/h}{k/h} := 0$ whenever k/h is not an integer in $\{0, \ldots, N/h\}$, and such a vanishing binomial annihilates its whole summand (so the accompanying factor with fractional exponent never occurs); all other binomial coefficients are the ordinary ones, and $0^0 := 1$ (relevant only at $\sigma = 1$, $k = N$).

In reference [1], this convention is also implicitly required by Eqs. (30)–(33) (Eq. (29) needs it only at $\sigma = 1$); in Eqs. (32)–(33) there, the factors $r(i)$ must be built from the primes r_i of $\gcd(k, n)$ rather than from the primes p_i of n (as printed, with the primes in increasing order, both equations give 36 at $n = 6$, $k = 3$, $\sigma = 2$, whereas $|S^a_{6^2,3}| = 48$).

Theorem 5.1 (k-active enumeration)**.** *For all integers $n \geq 2$, $d, \sigma \geq 1$, and $0 \leq k \leq N = n^d$, with $h(\mathbf{v}) = \prod_i p_i^{v_i}$, we have:*

$$|S^a_{n^d,k}| \;=\; \sum_{\substack{\mathbf{v} \in \{0,\ldots,d\}^{\omega(n)} \\ \mathbf{v} \neq \mathbf{0}}} (-1)^{1+|\mathbf{v}|} \prod_{i=1}^{\omega(n)} r_d(p_i, v_i) \binom{N/h(\mathbf{v})}{k/h(\mathbf{v})} (\sigma - 1)^{(N-k)/h(\mathbf{v})}.$$

Note: To avoid a notational conflict with the active-cell count k, the rank vectors here run over $\mathbf{v} \in \{0, \ldots, d\}^{\omega(n)}$, with one coordinate per distinct prime factor.

Proof. For a subgroup $H \leq \mathcal{A}$ with $|H| = h$, a configuration in $\mathrm{Fix}(H)$ is a free labeling of the N/h cosets; it has exactly k cells in state a iff exactly k/h cosets carry the label a — possible only when h divides k — and the remaining $(N-k)/h$ cosets carry labels from $\Sigma \setminus \{a\}$. Hence $|\mathrm{Fix}_k(H)| = \binom{N/h}{k/h}(\sigma-1)^{(N-k)/h}$, which is 0 under the stated convention unless $h \mid k$. The inclusion–exclusion and rank-vector grouping of Theorem 4.4 apply verbatim, since the summand again depends on J only through $|\langle \bigcup J \rangle| = h(\mathbf{v}(J))$. □

The theorem was verified exactly against definition-level brute force ($n = 2$, $d \in \{3, 4\}$, $\sigma \leq 3$, all k) and against the independent subgroup-level evaluator ($n \leq 6$ at $d = 3$; $n \leq 12$ at $d = 2$; all k), including the consistency check $\sum_k |S^a_{n^d,k}| = |S_{n^d}|$.

6 Bounds

We now turn our attention to lower bounds and upper bounds for the number of shift-symmetric configurations, with particular care regarding cases of equality.

6.1 Lower bound, attained exactly at primes

For the statement of our lower bound, we will evaluate Gaussian binomials at the (possibly composite) integer argument $q = n$; since $\begin{bmatrix} d \\ r \end{bmatrix}_q$ is a polynomial in q with nonnegative integer coefficients, all quantities are integers.

Theorem 6.1 (Lower bound; tight iff n prime)**.** *For all integers $n \geq 2$, $d \geq 1$, $\sigma \geq 2$:*

$$|S_{n^d}| \;\geq\; F_d(n,\sigma) \;:=\; \sum_{r=1}^{d} (-1)^{r+1}\, n^{\binom{r}{2}} \begin{bmatrix} d \\ r \end{bmatrix}_n \sigma^{\,n^{d-r}},$$

with equality if and only if n is prime.

The proof of Theorem 6.1, involving a monotone truncation chain $q^1 \leq q^2 \leq \cdots \leq q^k = |S_{n^d}|$ over the prime factors, is given in Appendix A. The argument makes use of an interpretation of the truncated sums as counts of unions of fixed-point sets on *rectangular* tori, plus a strict base-case inequality.

For $d = 2$, the statement exactly matches [1, Lemma E.21], and our chain lemma re-proves the corresponding step of their argument uniformly in n and σ.[3] A machine-checked proof of the theorem, taking a different and shorter route that bypasses the truncation chain altogether, is described in Section 11.

Remark 6.2 (The naive generalization is false)*.* A direct generalization of the two-term 2D bound, which would naively predict that $|S_{n^d}| \geq \sigma^{n^{d-1}\frac{n^d-1}{n-1}} - \sigma\big(\frac{n^d-1}{n-1} - 1\big)$ with equality iff n prime, is *false* for every $d \geq 3$. Indeed, at $(n, d, \sigma) = (2, 3, 2)$ it claims $|S_{2^3}| \geq 100$, while $|S_{2^3}| = 72$. For prime n, the true count is the full d-term alternating sum of Corollary 4.5, which agrees with the naive two-term expression only for $d \leq 2$ (when $\sigma \geq 2$).

Theorem 6.1 makes precise, and proves, the sense in which *primes minimize shift-symmetry in every dimension*: the count $|S_{n^d}|$ meets the universal lower envelope $F_d(n, \sigma)$ exactly at the primes and exceeds it strictly at every composite n.

6.2 Upper bound

We now derive two upper bounds on the number of shift-symmetric configurations.

Theorem 6.3 (Upper bounds)**.** *For all integers $n \geq 2$, $d \geq 1$, $\sigma \geq 2$:*

(i) (union form) $\quad |S_{n^d}| \;\leq\; \sum_{p \mid n} \begin{bmatrix} d \\ 1 \end{bmatrix}_p \sigma^{\,n^d/p}$;

(ii) (closed form) $\quad |S_{n^d}| \;\leq\; \big(2^d - 1 + 2\log_2 n\big)\, \sigma^{\,n^d/2}$.

Proof. (i). This is the union bound over Lemma 3.4, using $|\operatorname{Fix}(C)| = \sigma^{N/p}$ for $C \in \mathcal{L}_p$, as in the proof of [2, Thm. 5(ii)].
(ii). If $2 \mid n$, the $p = 2$ term of (i) is $(2^d - 1)\sigma^{N/2}$ (if n is odd, there is no such term and only the bound below is needed). We claim every *odd* prime $p \mid n$ contributes at most $2\sigma^{N/2}$; since n has at most $\log_2 n$ prime factors, (ii) follows. The claim, term by term: we must show $\begin{bmatrix} d \\ 1 \end{bmatrix}_p \leq 2\,\sigma^{N(\frac{1}{2}-\frac{1}{p})}$, and since $\sigma \geq 2$ and $\frac{1}{2} - \frac{1}{p} \geq \frac{1}{6}$ it suffices that $\begin{bmatrix} d \\ 1 \end{bmatrix}_p \leq 2^{1+N/6}$. For $d = 1$: $\begin{bmatrix} 1 \\ 1 \end{bmatrix}_p = 1$. For $d = 2$: any odd prime $p \mid n$ satisfies $p \leq n$, so $\begin{bmatrix} 2 \\ 1 \end{bmatrix}_p = p + 1 \leq n + 1$, and $n + 1 \leq 2^{1+n^2/6}$ for every $n \geq 3$: directly for $n \in \{3, 4, 5\}$ ($4 \leq 2^{2.5}$, $5 \leq 2^{3.67}$, $6 \leq 2^{5.17}$), and for $n \geq 6$ from $n^2/6 \geq n$ and $2^n \geq n + 1$. For $d \geq 3$: $\begin{bmatrix} d \\ 1 \end{bmatrix}_p \leq 2p^{d-1} \leq 2n^{d-1}$ and $N/6 \geq n^{d-1} \cdot n/6 \geq n^{d-1}/2$, so it suffices to show that $2^{\,n^{d-1}/2} \geq n^{d-1}$, which is true because $x = n^{d-1} \geq 9$ and $2^{x/2} \geq x$ for $x \geq 4$. □

Both parts were verified exactly for $n \leq 60$ and $d, \sigma \leq 4$. For comparison, on $G = \mathbb{Z}_n^d$ the general-group bounds give $|S| \leq \sigma^{N-1}$ [5, Cor. 1.7.2] and $|S| \leq (N-1)\sigma^{N/p_1}$ [2, Thm. 5(ii)],

[3] The published inequality chains behind the 2D bounds have small-n gaps (at $n \in \{6, 10, 14\}$); a companion erratum note repairs them uniformly. The d-dimensional chain of Appendix A has no exceptional lattice sizes.

p_1 the smallest prime factor. Bound (i) implies $|S| \le 2\log_2(n)\, n^{d-1}\, \sigma^{N/p_1}$ (Proposition 10.1), sharpening the prefactor $N-1 = n^d - 1$ of [2] at the same exponent for every $n \ge 7$ (and for $n \ge 5$ when $d \ge 2$; at $n = 2$ the union prefactor $2^d - 1$ exactly ties $N-1$, and for $n \in \{3, 4\}$ the two are comparable); for even $n \ge 4$ and $d \ge 2$, the closed form (ii) improves the prefactor $2\log_2(n)\, n^{d-1}$ further to $2^d - 1 + 2\log_2 n$. (For odd n the exponent $N/p_1 \le N/3$ of (i) is stronger than the $N/2$ of (ii); the two forms serve different purposes.) For $d = 2$ and $n \ge 4$, (ii) also slightly sharpens the $6\log_2(n)\sigma^{n^2/2}$ of [1, Lemma E.22].

7 Probability of shift-symmetry

In this section, we turn from questions of enumeration to the closely-related questions of probability. How likely is it for randomly selected configurations to possess non-trivial shift-symmetry?

Corollary 7.1 (Uniform distribution)**.** *For a configuration drawn uniformly from Σ^{n^d}, the probability $P^{\mathrm{unif}}_{n^d} = |S_{n^d}|/\sigma^{n^d}$ satisfies*

$$F_d(n,\sigma)\,\sigma^{-n^d} \;\le\; P^{\mathrm{unif}}_{n^d} \;\le\; \left(2^d - 1 + 2\log_2 n\right)\sigma^{-n^d/2},$$

where $F_d(n,\sigma)$ denotes the lower bound on $|S_{n^d}|$ given in Theorem 6.1. So $P^{\mathrm{unif}}_{n^d}$ decays as σ^{-cN} with $N = n^d$ and decay rate $c = \log_\sigma(1/P^{\mathrm{unif}}_{n^d})/N \in [\frac{1}{2} - o(1),\, 1 - \frac{1}{p_1}]$, where p_1 is the smallest prime factor of n. The lower end is given by Theorem 6.3(ii), and the upper end is the single-subgroup inclusion $\mathrm{Fix}(C) \subseteq S_{n^d}$ for one order-p_1 cyclic subgroup C, which gives $P^{\mathrm{unif}}_{n^d} \ge \sigma^{N/p_1 - N}$ (this is also the lower bound of Proposition 10.1). Here $o(1)$ refers to any regime with $N = n^d \to \infty$.

Corollary 7.2 (Density-uniform distribution)**.** *Let $\sigma \ge 2$. Under the density-uniform distribution (first draw $k \in \{0, \ldots, N\}$ uniformly, then a configuration with exactly k cells in state a uniformly), the probability that the drawn configuration lies in S_{n^d} is*

$$P^{\mathrm{dens}}_{n^d} \;=\; \frac{1}{N+1}\sum_{k=0}^{N} \frac{|S^a_{n^d,k}|}{\binom{N}{k}(\sigma-1)^{N-k}},$$

with $|S^a_{n^d,k}|$ given by Theorem 5.1.

Computational Observation 7.3 (Extremal pattern in n: true for $d = 2$, false for $d \ge 3$). For $d = 2$, $\sigma = 2$, exact computation for $n \le 41$ confirms the pattern of [1, Figs. 6, 8]: P^{unif} has local minima at every prime $5 \le n \le 40$ and local maxima at every even $4 < n \le 40$ (the boundary is a genuine exception: $P(3) = 26/2^9 > 736/2^{16} = P(4)$, so 3 is not a local minimum). For $d \ge 3$ the pattern *breaks*: at $d = 3$, $\sigma = 2$,

$$\log_{10} P(5) = -28.61 \;>\; \log_{10} P(6) = -31.67,$$

so the prime 5 is not a local minimum and the composite 6 is not a local maximum; likewise the even $n = 10$ is not a local maximum ($P^{\mathrm{unif}}_{9^3} > P^{\mathrm{unif}}_{10^3}$), and all three failures persist at $d = 4$. See Figure 2. This is consistent with the inversion law (Proposition 10.2): for the pair $(5, 6)$ of the pattern, $d_c(5) \approx 2.58$, and indeed $P^{\mathrm{unif}}_{5^d} > P^{\mathrm{unif}}_{6^d}$ for every $d \ge 3$ ($\sigma = 2$), which already breaks the pattern (as does $P^{\mathrm{unif}}_{9^3} > P^{\mathrm{unif}}_{10^3}$), while at $d = 3$ ($\sigma = 2$) no pair $(p, p \pm 1)$ with $7 \le p \le 23$ is inverted. Apart from two statements that hold for all $d \ge 3$ ($\sigma = 2$), resp. for all primes $p \ge 5$ and all $\sigma \ge 2$ — the inequality $P^{\mathrm{unif}}_{5^d} > P^{\mathrm{unif}}_{6^d}$ above, and, at $d = 2$, $P^{\mathrm{unif}}_{p^2} < P^{\mathrm{unif}}_{(p\pm1)^2}$ (Observation 10.5) — these are exact finite computations (all machine-checked, Section 11), not asymptotic claims; we know of no proof of the even-peak pattern as a theorem for general n.

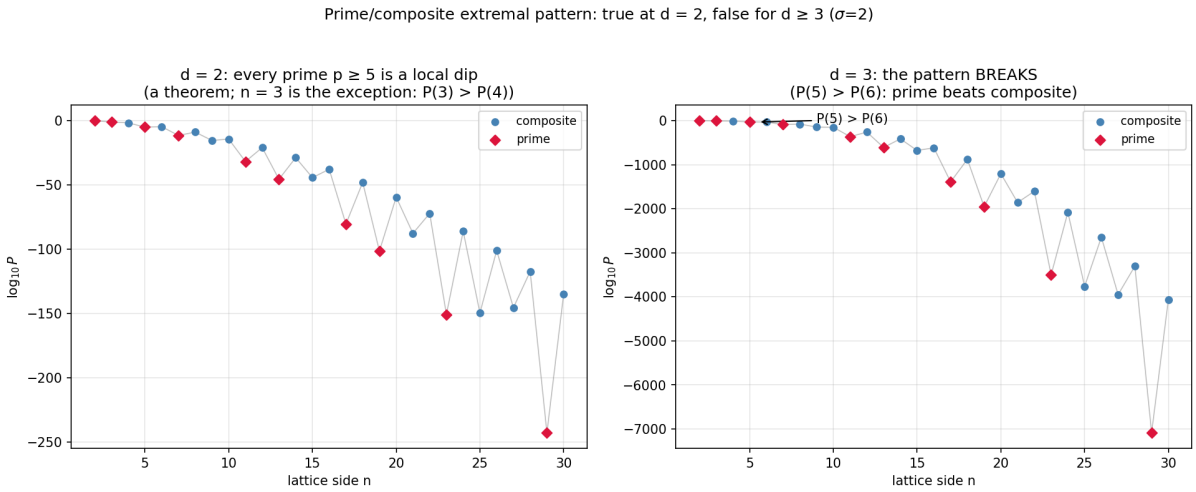


Figure 2: The prime/composite extremal pattern of Observation 7.3 ($\sigma = 2$; the exact formula, evaluated numerically). Left, $d = 2$: $\log_{10} P^{\mathrm{unif}}$ has local dips at every prime $5 \le n \le 40$ (shown here for $n \le 30$); the boundary case $n = 3$ is the audited exception. Right, $d = 3$: the pattern breaks — already $P(5^3) > P(6^3)$, so the prime 5 is no longer a local minimum.

8 Dynamics: irreversibility, insolvability, rank monotonicity

In this section we establish the dynamical consequences of shift-symmetry in configurations.

Theorem 8.1 (Irreversibility)**.** *Let Φ be translation-equivariant (Definition 2.4). Then, for every configuration* $\mathbf{s}$, $\mathrm{Stab}(\Phi(\mathbf{s})) \supseteq \mathrm{Stab}(\mathbf{s})$*; in particular,* $\mathbf{s} \in S_{n^d}(\mathbf{v})$ *implies* $\Phi(\mathbf{s}) \in S_{n^d}(\mathbf{v})$*, and* $\Phi(S_{n^d}) \subseteq S_{n^d}$*.*

Proof. If $\tau_{\mathbf{v}}\mathbf{s} = \mathbf{s}$, then $\tau_{\mathbf{v}}\Phi(\mathbf{s}) = \Phi(\tau_{\mathbf{v}}\mathbf{s}) = \Phi(\mathbf{s})$. □

For cellular automata on any finite group, this inclusion appears in the proof of Lemma 10 of [2], whose statement gives the converse for distinct orbits; see also [3, Thm. 2(iii)]. By Remark 2.5, the equivariance hypothesis cannot be weakened to the assumption "all cells share the local rule."

Corollary 8.2 (Insolvability)**.** *Let $\sigma \ge 2$ and $N > 1$. Any task requiring, on some input, an output configuration outside S_{n^d} reachable from an input inside S_{n^d} is unsolvable by translation-equivariant CA on $\mathbb{Z}_n^d$. Indeed, symmetric inputs exist (all constants lie in S_{n^d}), and by Theorem 8.1 no trajectory leaves S_{n^d}. This covers leader election (the target configuration with a unique distinguished cell is asymmetric for $N > 1$); d-dimensional leader-election algorithms as in [17, 18] therefore necessarily break one of the hypotheses (uniformity, determinism, synchrony) or restrict the admissible inputs, e.g. to asymmetric configurations. More generally, since* Stab *can only grow along a trajectory (Theorem 8.1), every pattern-recognition or image-processing task whose required output breaks a symmetry of some admissible input is unsolvable, as in the two-dimensional discussion of [1].*

Proposition 8.3 (Per-prime rank monotonicity)**.** *For a prime $p \mid n$ define the* p-rank *of* $\mathbf{s}$ *as* $\mathrm{rk}_p(\mathbf{s}) = \dim_{\mathbb{F}_p}\big(\mathrm{Stab}(\mathbf{s}) \cap \mathcal{A}[p]\big) \in \{0, \dots, d\}$ *(via Lemma 3.3). Under any translation-equivariant Φ, every p-rank is non-decreasing along every trajectory:* $\mathrm{rk}_p(\Phi(\mathbf{s})) \ge \mathrm{rk}_p(\mathbf{s})$ *for all $p \mid n$ simultaneously.*

Proof. By Theorem 8.1, $\mathrm{Stab}(\Phi(\mathbf{s})) \cap \mathcal{A}[p] \supseteq \mathrm{Stab}(\mathbf{s}) \cap \mathcal{A}[p]$, and both are $\mathbb{F}_p$-subspaces. □

For composite n there is one rank per prime factor, and all of them are monotone. Geometrically, $\mathrm{rk}_p = 1$ configurations are periodic along one direction of the p-torsion ("stripes"/"slabs"), $\mathrm{rk}_p = d$ configurations are exactly those fixed by the whole p-torsion $\mathcal{A}[p] = (\frac{n}{p}\mathbb{Z}/n\mathbb{Z})^d$; each superlevel set $\{\mathrm{rk}_p \geq r\}$ is absorbing (no trajectory leaves it) by Proposition 8.3. Kornyak's observation that deterministic dynamics cannot decrease symmetry [16] is the general heuristic; the proposition is its exact translation-stabilizer form.

9 Detection

In this section, we discuss how to determine whether configurations possess shift-symmetry.

The detector given here generalizes that given in an earlier paper [1, Sec. V]. Assume the prime factorization of n is given (otherwise add one factorization of an integer of $\log_2 n$ bits), and count cell-read operations; d is part of the input.

> **Algorithm.** For each prime $p \mid n$ and each of the $\begin{bmatrix} d \\ 1 \end{bmatrix}_p$ order-p cyclic subgroups $C = \langle \mathbf{w} \rangle$ (enumerated via one representative line each, Lemma 3.3): traverse the cells in a fixed order (which may depend on the subgroup); for each yet-unvisited cell, walk its C-orbit (p cells) checking equality; abort the subgroup at the first failing orbit. Report "symmetric" at the first subgroup all of whose orbits pass; report "asymmetric" if all subgroups fail.

Correctness is validated by Lemma 3.4: a configuration is shift-symmetric iff it is fixed by some order-p cyclic subgroup with $p \mid n$.

Throughout this section we charge one read per visited cell: the first cell of an orbit walk (the *anchor*) is read once and its value cached, and each subsequent cell of the orbit is read once and compared to the cached value. A failing orbit walk may stop at its first mismatch or be completed; all results of this section hold in both variants, except the exact factor in Remark 9.3, which refers to completed walks.

In what follows, let $\mathrm{sopf}_d(n) = \sum_{p|n} \begin{bmatrix} d \\ 1 \end{bmatrix}_p = \sum_{p|n} \frac{p^d-1}{p-1}$ denote the total number of order-p cyclic subgroups over all primes $p \mid n$ (the d-dimensional analogue of the integer logarithm).

Theorem 9.1 (Worst case)**.** *The algorithm reads at most $N \cdot \mathrm{sopf}_d(n) = O(\mathrm{sopf}_d(n) \cdot n^d)$ cells.*

Proof. Per subgroup, an orbit walk of length p performs p reads, and the walked orbits are disjoint, so one subgroup costs at most N reads; there are $\mathrm{sopf}_d(n)$ subgroups. □

Proposition 9.2 (Average case, uniform random input)**.** *For $\mathbf{s}$ uniform on Σ^{n^d}, the test of any single order-p subgroup, run on its own, visits in expectation exactly $\frac{1-Q^m}{1-Q}$ orbits, with $Q = \sigma^{1-p}$ and $m = N/p$. Consequently, the expected total number of cells read by the full algorithm is at most*

$$\sum_{p|n} \begin{bmatrix} d \\ 1 \end{bmatrix}_p \frac{p}{1-\sigma^{1-p}} \;\leq\; 2 \sum_{p|n} p \begin{bmatrix} d \\ 1 \end{bmatrix}_p \;=\; O\Big(\sum_{p|n} p^{\,d}\Big).$$

Proof. Orbits of a fixed subgroup partition the cells, so under the product measure the events "orbit is monochromatic" are independent, each of probability σ^{1-p}; the walk visits i orbits iff the first $i-1$ pass, giving the truncated geometric expectation $\sum_{i=0}^{m-1} Q^i$. Summing $p \cdot (1-Q)^{-1}$ over all subgroups bounds the total; $(1-\sigma^{1-p})^{-1} \leq 2$ for $\sigma, p \geq 2$. □

The per-generator expectation was confirmed to match full-enumeration averages *exactly* for $(n, d, \sigma) \in \{(2,3,2), (2,3,3), (2,4,2)\}$ and, in two dimensions, for $n \leq 4$ with $\sigma = 2$ and for $(n, \sigma) = (3, 3)$.

Remark 9.3 (On "polylogarithmic" average cases). $\sum_{p|n} p^d$ is *not* polylogarithmic in N in general: for prime n it equals $n^d = N$. The polylogarithmic figures quoted in [1, Sec. V–VI] for $d = 2$ are stated there for *randomly chosen* n and rest on $\sum_{p|n} p$ and $\sum_{p|n} p^2$ being polylogarithmic; but for each fixed dimension d, $\sum_{p|n} p^d$ is polylogarithmic in N (with a degree depending on d) when all prime factors of n are polylogarithmically small, whereas for any fixed C and j, the n with $\sum_{p|n} p^d \le C \log^j N$ have natural density zero, so this bound is not polylogarithmic for almost all n. Note also that, for prime n, the average case $O(N)$ still improves the worst case, which is of exact order $N \cdot n^{d-1}$, by a factor of at least order n^{d-1} (of exact order n^{d-1} when every failing orbit walk is completed, as in the proofs of Theorem 9.1 and Proposition 9.2).

Remark 9.4 (Read complexity). With memory for all N cells, N reads suffice: read every cell once and test in memory. For $N \ge 2$ and $\sigma \ge 2$, N reads are also necessary in the worst case: every correct deterministic algorithm, adaptive or not, makes at least N reads on every constant configuration. Indeed, a constant configuration is shift-symmetric, while the configuration that differs from it in a single unread cell has trivial stabilizer and yields the same reads, hence the same answer. Writing I_a for the indicator of the cells in state a, a translation $\mathbf{w}$ lies in $\mathrm{Stab}(\mathbf{s})$ if and only if $\sum_{a\in\Sigma} \sum_{\mathbf{u}\in\mathcal{A}} I_a(\mathbf{u})\, I_a(\mathbf{u}+\mathbf{w}) = N$.

If the configuration is held in memory, the sums of Remark 9.4 are obtained for all $\mathbf{w}$ at once by d-dimensional fast Fourier transforms [20, 21], one per symbol that occurs in $\mathbf{s}$, in $O(\min(\sigma, N) \cdot N \log N)$ arithmetic operations, without rounding error if a number-theoretic transform is used [22]. This improves on the worst case of Theorem 9.1 when $\mathrm{sopf}_d(n)$ is large compared with $\min(\sigma, N) \log N$; for fixed σ, this is the case for $d \ge 2$ as the prime n grows, and for $n = 2$ as d grows. For fixed d and n a power of 2, for $d = 1$, and on average (Proposition 9.2), the orbit walks are cheaper.

Related decision problems on toroidal arrays — shift-equivalence of tilings and recognition of toroidal codes, including undecidability results — were recently analyzed by Anselmo, Cavallaro, Madonia, and Selmi [19]; detecting a nontrivial translation stabilizer, as here, is the tractable member of that family.

10 Crossover laws and empirical observations in higher dimensions

We first prove two crossover laws that organize all the observations made in this section; the observations after them are computational findings with exact arithmetic, labeled as such. Where an item after the two laws states an exact claim for all parameters (such as the composite advantage at $d = 2$ and the deepening of the inversion in Observation 10.5), that claim is a theorem, machine-checked in Section 11.

Proposition 10.1 (Decay-rate law: only the smallest prime factor survives)**.** *Let $p_1 = p_1(n)$ be the smallest prime factor of n and $N = n^d$. For all $n \ge 2$, $d \ge 1$, $\sigma \ge 2$:*

$$\sigma^{N/p_1} \;\le\; |S_{n^d}| \;\le\; 2\log_2(n)\, n^{d-1}\, \sigma^{N/p_1}.$$

Consequently the decay rate $c(n,d) := -\log_\sigma P^{\mathrm{unif}}_{n^d}/N$ satisfies

$$0 \;\le\; \Big(1 - \frac{1}{p_1}\Big) - c(n,d) \;\le\; \frac{\log_\sigma\big(2\, n^{d-1} \log_2 n\big)}{N} \;\xrightarrow[d\to\infty]{}\; 0,$$

so $c(n,d) \to 1 - 1/p_1(n)$. Asymptotically in d, lattice sides fall into universality classes *determined solely by their smallest prime factor, with even n on the extremal class $c = 1/2$ (the floor permitted by Theorem 6.3) and primes p on $c = 1 - 1/p$.*

Proof. Lower bound: $\mathrm{Fix}(C) \subseteq S_{n^d}$ for any single order-p_1 cyclic subgroup C, and $|\mathrm{Fix}(C)| = \sigma^{N/p_1}$ (Lemma 3.1). Upper bound: in Theorem 6.3(i), every exponent satisfies $N/p \le N/p_1$, and $\sum_{p|n} \begin{bmatrix} d \\ 1 \end{bmatrix}_p \le \sum_{p|n} 2p^{d-1} \le 2\,\omega(n)\,n^{d-1} \le 2\log_2(n)\,n^{d-1}$. The prefactor $N-1$ of [2, Thm. 5(ii)] gives the same limit. □

Figure 3 shows the convergence in numerical evaluations of the exact formula.

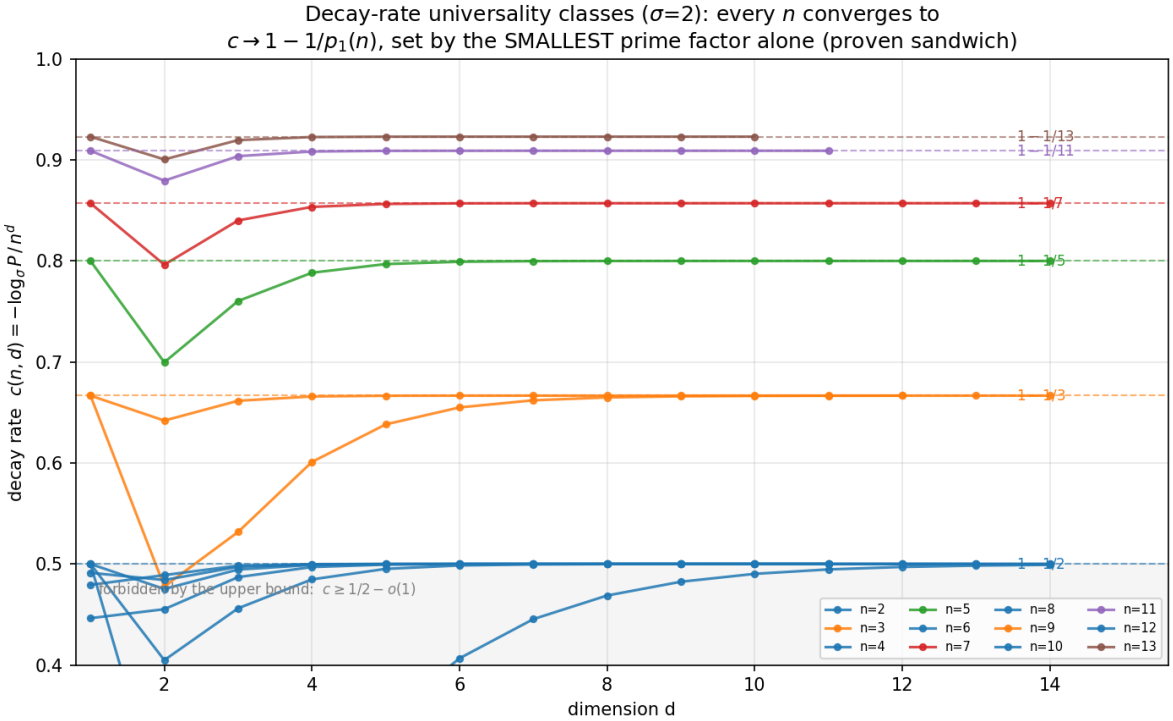


Figure 3: Decay-rate universality classes (Proposition 10.1): the measured rate $c(n,d) = -\log_\sigma P_{n^d}^{\mathrm{unif}}/n^d$ at $\sigma = 2$ for $2 \le n \le 13$, plotted against d (the exact formula, evaluated numerically). Every lattice side converges to the level $1 - 1/p_1(n)$ set by its smallest prime factor alone; all even n collapse onto the extremal class $c = 1/2$, the floor permitted by Theorem 6.3 (shaded region excluded asymptotically).

Proposition 10.2 (Inversion law)**.** *Let $p \ge 3$ be prime, $\sigma \ge 2$, and set $\varphi(d) = (1 - \frac{1}{p})\,p^d - \frac{1}{2}(p+1)^d$. Then:*

(i) if $\varphi(d) > \log_\sigma\big(2\log_2(p)\,p^{d-1}\big)$, then $P_{(p+1)^d}^{\mathrm{unif}} > P_{p^d}^{\mathrm{unif}}$ (composite advantage);

(ii) if $\varphi(d) < -\log_\sigma\big(2\log_2(p+1)\,(p+1)^{d-1}\big)$, then $P_{(p+1)^d}^{\mathrm{unif}} < P_{p^d}^{\mathrm{unif}}$ (prime advantage).

The function φ, regarded as a function of real d, is positive below and negative above its unique zero

$$d_c(p) \;=\; \frac{\ln\big(2 - \frac{2}{p}\big)}{\ln\big(1 + \frac{1}{p}\big)} \;=\; p\ln 2 + O(1) \qquad (p \to \infty),$$

so the two-dimensional "composite advantage" inverts within an explicitly bounded window around $d_c(p)$, a dimension that grows linearly *in the prime: (i) applies to every integer $d \ge 1$ with $d \le d_c(p) - 20$, and (ii) to every integer $d \ge d_c(p) + 20$.*

Proof. (i) By Proposition 10.1 applied to $n = p+1$ (whose smallest prime factor is 2) and to $n = p$: $P_{(p+1)^d} \ge \sigma^{-(p+1)^d/2}$ and $P_{p^d} \le 2\log_2(p)\,p^{d-1}\sigma^{-p^d(1-1/p)}$, so $\log_\sigma\big(P_{(p+1)^d}/P_{p^d}\big) \ge \varphi(d) -$

$\log_\sigma(2\log_2(p)p^{d-1}) > 0$. (ii) is symmetric. Monotonicity of $d \mapsto \varphi(d)/p^d = (1-\frac{1}{p}) - \frac{1}{2}(1+\frac{1}{p})^d$ gives the unique sign change at $d_c(p)$, and $\ln(2-2/p)/\ln(1+1/p) = p\ln 2 + O(1)$ by expanding both logarithms. The window bound follows by elementary estimates of φ and of the thresholds in (i) and (ii); it is machine-checked (Section 11), and the constant 20 is not optimized. □

Remark 10.3 (Measured inversion points). Linearly interpolated zeros of $d \mapsto \log_{10}[P_{(p+1)^d}/P_{p^d}]$ between the consecutive integer dimensions at its first sign change, in exact data ($\sigma = 2$), against $d_c(p)$:

p	3	5	7	11	13
measured	1.00	2.10	4.00	6.37	8.03
$d_c(p)$	1.00	2.58	4.04	6.87	8.27

Measured values sit at or below d_c, mainly because of the interpolation itself: interpolating the bare exponent φ (without coefficient corrections) linearly between $\lfloor d_c \rfloor$ and $\lfloor d_c \rfloor + 1$ already gives a zero below $d_c(p)$ for $p = 5, 7, 11, 13$ (at $p = 3$ both equal 1); the coefficient corrections (the factors $|S_{n^d}|/\sigma^{N/p_1}$, which Proposition 10.1 bounds on both sides) favor the prime side but account for at most a quarter of the gap (at most 10^{-3} of it for $p = 11, 13$). At $p = 3$ the boundary case is exact and alphabet-independent: $P_3^{\text{unif}} = P_4^{\text{unif}} = \sigma^{-2}$ at $d = 1$, since $|S_3| = \sigma$ and $|S_4| = \sigma^2$. Figure 4 displays both the sign flips and the measured-versus-predicted inversion dimensions.

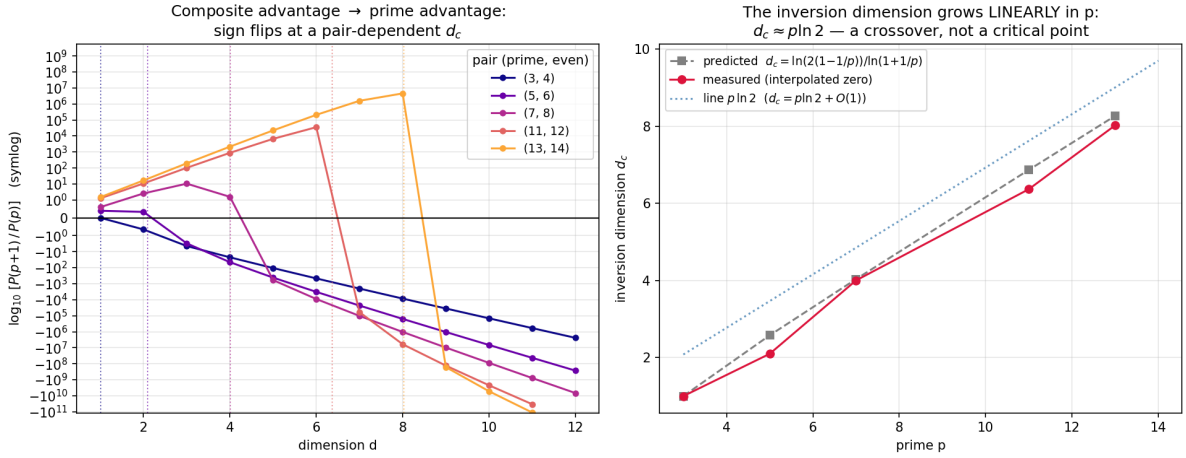


Figure 4: The inversion law (Proposition 10.2, Remark 10.3; $\sigma = 2$; the exact formula, evaluated numerically). Left: the ratio $\log_{10}\big[P^{\text{unif}}_{(p+1)^d}/P^{\text{unif}}_{p^d}\big]$ for the prime–even pairs $(p, p+1)$, $p \in \{3, 5, 7, 11, 13\}$, flips sign at a pair-dependent inversion dimension (symlog scale; dotted verticals mark the interpolated zeros). Right: the measured inversion dimensions against the prediction $d_c(p) = \ln(2 - 2/p)/\ln(1 + 1/p)$ and the line $p\ln 2$ ($d_c(p) = p\ln 2 + O(1)$; the linear asymptote itself is $p\ln 2 + \frac{1}{2}\ln 2 - 1$).

The remaining observations are computational, except for exact claims stated for all parameters (such as the composite advantage at $d = 2$ and the deepening of the inversion in Observation 10.5), which are theorems (machine-checked, Section 11).

Computational Observation 10.4 (Single-term crossover dimension). For prime $n = p$, define $d^*(p; \theta)$ as the least $d \geq 2$ at which the second term of Corollary 4.5 falls below a fraction θ of the first, i.e. $|T_2/T_1| < \theta$ where $|T_2/T_1| = p\begin{bmatrix} d \\ 2 \end{bmatrix}_p \begin{bmatrix} d \\ 1 \end{bmatrix}_p^{-1} \sigma^{-p^{d-2}(p-1)}$. Computing d^* for all 95 primes up to 499 ($\sigma = 2$): only $p \in \{2, 3, 5\}$ have $d^* > 2$ at $\theta = 0.05$; a fit $d^* \approx 1 + C/\ln p$ to a fractional version of d^* (interpolating $\log|T_2/T_1|$ linearly between consecutive integer dimensions) gives

$C \approx 2.2$–2.65 depending on θ (about 2.5 ± 0.3), but with only two to four unclamped data points per threshold (for the integer d^*: four, three, and two at $\theta = 0.01, 0.05, 0.10$) and systematic residuals this is an indicative, threshold-dependent summary, not an estimate of a universal constant. The driver is the doubly-exponential suppression $\sigma^{-p^{d-2}(p-1)}$.

Computational Observation 10.5 (Prime–composite gap inversion). At $d = 2$, for every $\sigma \geq 2$, every prime $p \geq 5$ has smaller P^{unif} than both adjacent integers $p \pm 1$ (a theorem, machine-checked in Section 11; the pair $(3, 4)$ is the exception: $P^{\text{unif}}_{4^2} < P^{\text{unif}}_{3^2}$; cf. Observation 7.3); by $d = 3$ ($\sigma = 2$) this inverts for the pair $(5, 6)$ (Observation 7.3), but for no pair $(p, p \pm 1)$ with $7 \leq p \leq 23$, and the inversion deepens with d: for every prime $p \geq 3$ and every $\sigma \geq 2$, once $P^{\text{unif}}_{(p+1)^d} < P^{\text{unif}}_{p^d}$, the ratio $P^{\text{unif}}_{(p+1)^d}/P^{\text{unif}}_{p^d}$ decreases strictly with every further dimension (a theorem, machine-checked in Section 11; e.g., $P(6)/P(5)$ falls from $\sim 10^{0.33}$ at $d = 2$ to $10^{-419.2}$ at $d = 5$, $\sigma = 2$).

Remark 10.6 (Duality). $\begin{bmatrix} d \\ r \end{bmatrix}_p = \begin{bmatrix} d \\ d-r \end{bmatrix}_p$ for $0 \leq r \leq d$, so $\mathbb{F}_p^d$ has equally many subspaces of dimensions r and $d - r$ (equivalently, for a prime $p \mid n$, $\mathcal{A}[p]$ has equally many subgroups of orders p^r and p^{d-r}). We state this purely as the standard symmetry of Gaussian binomials; we do not attach further structural claims to it.

Reproducibility. All counts and verification computations use exact integers; the figures evaluate the exact formula numerically in log space (cross-checked against exact integers where feasible). The verification suite (definition-level brute force; independent subgroup-level evaluators for $|S|$ and $|S^a_k|$; bound and extremal-pattern scans; the detection-expectation enumeration) and the chart-generation scripts (`compute_charts.py`, `compute_criticality.py`, `compute_C_exploration.py`, `compute_dimension_insights.py`, which produces Figures 2–4, and `compute_example_configs.py`, which produces Figure 1 and checks its stabilizers and count) are in the project repository. Commands, environments, outputs, and SHA-256 checksums are recorded in `research/experiments/README.md`.

11 Formal verification

The mathematical content of every numbered statement of Sections 2–10 and Appendices A and B has been formally verified in the Lean 4 proof assistant (v4.33.0-rc1) on top of the mathlib library: that of all definitions, lemmas, theorems, propositions, corollaries, remarks, and computational observations there, of Table 1, of the table of Remark 10.3, and of the captions of Figures 2–4, together with every mathematical claim in the surrounding (non-proof) prose that is precise enough to be stated formally. Statistical summaries (such as the fitted constant C of Observation 10.4), vague, bibliographic, and record-keeping sentences, comparisons of operation counts that rest on cited results (after Remark 9.4), and the summary table of the concluding section are not formalized. Where the verification found a mathematical claim ambiguous or false, the text above states the corrected claim, and that is the claim certified. The illustrative Figures 1 and 5 and the applications discussed in Section 12 are not formalized; the stabilizers in the caption of Figure 1 were checked by exact computation, and its count is one of the machine-checked reference values of Section 4.4. Figure 5 summarizes the verification.

The development has two layers. The core layer (≈3200 lines: 15 lemma modules plus definition, statement, experiment, and audit files) defines the model from first principles — lattice, configurations, shifts, stabilizers, fixed sets, equivariant maps — and proves the two main theorems:

- `enumeration_proved`: Theorem 4.4 for all $n \geq 2$, $d \geq 1$, $\sigma \geq 1$, exactly as stated ($\sigma \geq 1$ being weaker than the standing alphabet assumption $\sigma \geq 2$ of Section 2);

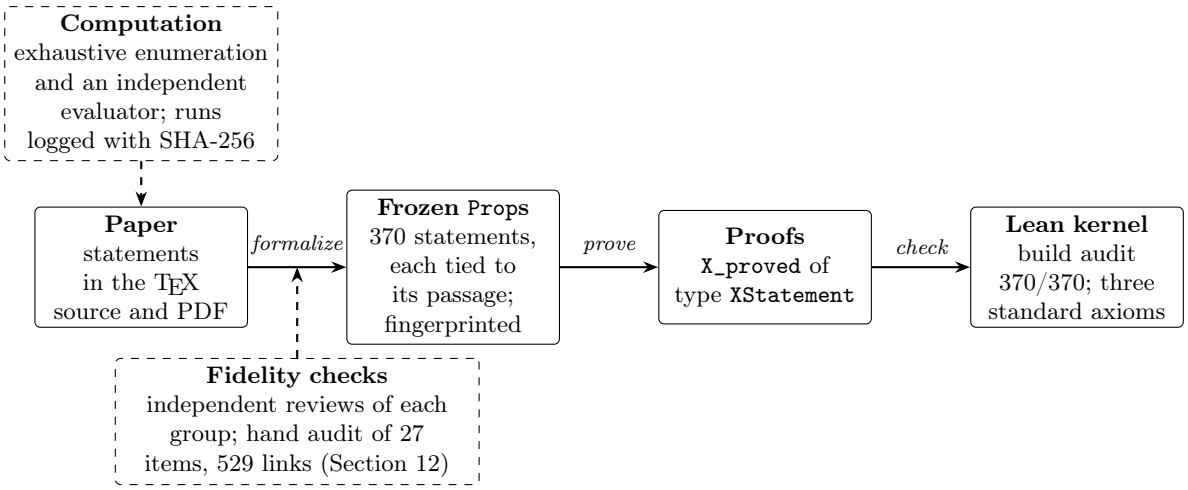


Figure 5: The verification pipeline. Statements of the paper are frozen as Lean `Prop`s (all but five before their proofs were written); each proof `X_proved` must have exactly the frozen type; the Lean kernel checks the proofs, and an audit module in the build fails unless every frozen statement (currently 370) is proved from the three standard axioms. Whether a frozen statement says what the paper says is checked separately (dashed), by independent fidelity reviews and, for the theorems, lemmas, propositions, and corollaries, by the hand audit described in Section 12; formulas and numbers are, in addition, checked by exact computation.

- `lowerBound_proved`: Theorem 6.1 verbatim ($n \geq 2$, $d \geq 1$, $\sigma \geq 2$), including both directions of the equality-iff-prime characterization.

The paper-wide layer (≈27 000 lines in 67 modules) states and proves everything else: 370 `Prop`-valued statements (including selected proof-internal claims, representation bridges, a few disclosed strengthenings, and machine-checked evidence for the corrections), organized by part of the paper — Section 2 with the convention sentence opening Section 3 (31), the rest of Section 3 (20), Lemmas 4.1–4.2, Definition 4.3, Table 1 and the Duality remark (22), the rest of Section 4 (48, and 5 more for the necklace identification in Remark 4.6(iii)), Section 4.5 with Appendix B (17), Section 5 with Corollary 7.2 (31), Section 6 with Corollary 7.1 (41), Section 8 with the caption of Figure 2 (33), Section 9 (31), Propositions 10.1–10.2 with the caption of Figure 3 (24), Appendix A (33), and Observation 7.3, Remark 10.3 with the caption of Figure 4, and Observations 10.4–10.5 (34). Some of its proofs also use certified results of a companion development (two further modules, ≈440 lines): the proof of Theorem 5.1 uses the relative enumeration theorem, which extends the inclusion–exclusion behind Theorem 4.4 from the set of all configurations to an arbitrary set of configurations, and proofs in Appendix A use its weight-generic regrouping lemma. Multi-part results are stated part by part and as a whole; asymptotic claims carry explicit quantifiers; a rounded decimal is stated as the interval its rounding asserts; the detection algorithm of Section 9 is a Lean function with explicit read counting, and its statements are quantified over every enumeration of the subgroups, every traversal order, and both variants of a failing orbit walk (except the exact factor of Remark 9.3, which is stated for completed walks only); the density-uniform distribution of Corollary 7.2 is a probability mass function in mathlib. Numerical claims are proved with exact integer certificates checked by the Lean kernel, combined, for large lattices (up to 14^9 cells in Remark 10.3), with proved analytic bounds such as the one-fixed-set and union bounds; no compiled code or floating-point arithmetic is trusted. Several of the 370 statements record, as machine-checked facts, why an earlier wording or a cited formula fails in its literal reading.

All formal statements were frozen as `Prop`-valued definitions — all but five of them *before* their proofs were written — each with a documentation string that locates the passage it formal-

izes, in most cases quoting it verbatim, and records its representation choices (e.g. the alphabet is realized as $\{0, \dots, \sigma-1\}$, harmless since only $|\Sigma|$ enters; rank vectors are reindexed by divisors of $(p_1 \cdots p_k)^d$; quotients are exact), and each group of statements passed independent reviews of its fidelity to the paper. In addition, the correspondence of every theorem, lemma, proposition, and corollary with its frozen statement is audited by hand (Section 12). Every frozen statement `XStatement` of the core and paper-wide layers has a theorem `X_proved` whose type is literally `XStatement` or, for a statement with parameters, $\forall$`params, XStatement params`. A few details of the corrected text are certified by auxiliary theorems of the formal proofs, which are kernel-checked but were not frozen in advance. One of these details, the constant 20 in Proposition 10.2, is also stated in three of the five exceptions above; the other two state the symmetry-breaking sentence of Corollary 8.2. All five were frozen after a proof of their content existed, and passed three independent reviews of their fidelity to the paper before their `X_proved` theorems were written. Two audit modules are part of the default build: one prints the axiom set of every declaration it lists from the core layer and from companion developments, and one discovers all frozen statements of the paper-wide layer from the environment and fails the build unless each has such a theorem depending only on the three standard axioms of the Lean/mathlib foundation (propositional extensionality, choice, quotient soundness). The development contains no unproved placeholder and no further axiom, and every theorem and every safe definition in the 89 modules reachable from the library's root module, the root included (3036 theorems and 804 definitions), was additionally re-checked by the Lean kernel.

Three features of the formalization deserve mention. First, it is self-contained precisely where the prose leans on the literature: the signed spanning-set count (Lemmas 4.1–4.2), the Grassmannian subspace count, and the p-group Möbius evaluation (in its subspace-lattice instance) are proved from first principles; so are the ingredients of the second derivation of Section 4.3 — Lemma 7(i) of [2] for every finite group, the product rule for the Möbius function of the subgroup lattice, and Hall's evaluation on the primary components of $\mathcal{A}$ — as well as the bounds of [5] and [2] specialized to $\mathbb{Z}_n^d$ and [1, Lemmas E.21, E.22]. No verified step rests on the cited results of Stanley, Hall, or Hawkes–Isaacs–Özaydin. Second, the certified proof of Theorem 6.1 takes a shorter route than Appendix A: with the enumeration theorem available as a verified identity, the prime case reduces to a reindexing of divisors of p^d, and the composite case follows from the single fixed-set bound $|S_{n^d}| \geq \sigma^{n^d/p_1}$ (p_1 the least prime factor) together with elementary coefficient estimates (chiefly $\begin{bmatrix} d \\ r \end{bmatrix}_q \leq 2^d q^{r(d-r)}$). In particular the certified chain nowhere invokes [1, Lemma E.21] or its erratum, so the verified results are independent of the published 2D bounds and their repairs. Third, and more generally, the prose proofs are not replayed line by line: the formal proofs follow their own routes, and proof-internal claims of the prose proofs are formalized only selectively.

The Lean sources reside in `research/formal/shiftsym/` in the project repository (the paper-wide layer in `Shiftsym/Paper/` and `Shiftsym/AuditPaper.lean`), together with the formalization plan and the statement-preservation table.

12 Summary and conclusion

The structure of shift-symmetry on $\mathbb{Z}_n^d$ is governed by the prime factorization of n through the subspace lattices of $\mathbb{F}_p^d$: the Gaussian binomial coefficients replace the ad hoc 2D coefficients, the Möbius values $(-1)^r p^{\binom{r}{2}}$ supply the signs, and every dynamical statement flows from translation equivariance alone. The p-torsion carries all of this: a configuration is shift-symmetric iff some subgroup of prime order fixes it (Lemma 3.4), the detector tests only these subgroups (Section 9), translation-equivariant dynamics never lower $\dim_{\mathbb{F}_p}(\mathrm{Stab}(\mathbf{s}) \cap \mathcal{A}[p])$ (Proposition 8.3), and on a finite abelian group the count depends only on its order and the dimensions of its p-torsion (Corollary B.1). Two cautionary findings of the verification deserve emphasis: naive dimensional lifting of true 2D statements can fail outright (Remark 6.2, Observation 7.3), and

Table 2: 2D ([1]) vs. the d-dimensional generalization.

Component	**2D**	**d-dimensional (this paper)**
Lattice, cells	$\mathbb{Z}_n^2$, $N = n^2$	$\mathbb{Z}_n^d$, $N = n^d$
Order-p subgroups per prime	$p + 1$	$(p^d - 1)/(p - 1)$
Coefficients	$1,\ p{+}1,\ p$	$r_d(p, v) = p^{\binom{v}{2}} \begin{bmatrix} d \\ v \end{bmatrix}_p$
Enumeration	Thm. E.19	Theorem 4.4; finite abelian groups: Theorem 4.7
k-active counts	Thm. E.32	Theorem 5.1
Lower bound (tight iff prime)	$(n{+}1)\sigma^n{-}n\sigma$	$F_d(n, \sigma)$, Theorem 6.1
Upper bound	$6\log_2(n)\sigma^{n^2/2}$	$(2^d{-}1{+}2\log_2 n)\sigma^{n^d/2}$
Irreversibility	Thm. E.5	Theorem 8.1 (equivariance explicit) + Proposition 8.3
Detection, worst case	$O(n^3)$	$O(\mathrm{sopf}_d(n) \cdot N)$
Detection, average case	$O(n^2)$; polylog claimed for random n	$O(\sum_{p\mid n} p^d)$; this bound is polylog (fixed d) only on a set of n of density 0 (e.g. polylog-smooth n)
Decay rate of $P = \sigma^{-cN}$	—	$c \to 1 - 1/p_1(n)$ (Proposition 10.1)
Prime local minima of P	yes for $p \geq 5$ (empirical there; a theorem here, Observation 10.5)	**fails for** $d \geq 3$ (at $p = 5$, $\sigma = 2$); inversion of $(p, p{+}1)$ around $d_c(p) \approx p \ln 2$ (Proposition 10.2, Observation 7.3)

the enumeration principle itself, though not the multi-prime closed form, was already available in the general group-theoretic literature [2] — both are reminders that the value added here lies in the explicit formulas, the tightness characterization, and the cellular-automata consequences, each verified computationally at the definition level.

Applications. The counts use only a finite alphabet and a torus, and the dynamical results use only translation equivariance (their proofs do not need a finite alphabet), so they apply wherever these ingredients occur.

- *Distributed computing.* A CA on $\mathbb{Z}_n^d$ is a synchronous anonymous network (a torus), a special case of the networks for which Angluin proved the classical impossibility of symmetry breaking [28]; Yamashita and Kameda characterized when leader election and related problems are solvable on anonymous networks [31]. For CA, the symmetry obstruction and its use in performance bounds were studied in one dimension by Banda, Caughman, and Pospíchal [30], and in two dimensions by Banda et al. [1]. Corollary 8.2 and the counts extend this obstruction and its quantitative form to arbitrary dimension: for $N > 1$, every translation-equivariant CA fails at leader election on all inputs in S_{n^d}, a fraction $P_{n^d}^{\mathrm{unif}}$ of all inputs, bounded in Corollary 7.1.

- *Machine learning.* A stride-one convolutional layer with circular padding, shared weights, and pointwise nonlinearities is a translation-equivariant map of feature maps on $\mathbb{Z}_n^d$ [32]. Since the proof of Theorem 8.1 uses equivariance alone, no stack of such layers can remove a translation symmetry of its input, an instance of the limitation of equivariant networks discussed by Kaba and Ravanbakhsh [33]. Neural cellular automata [34] grow patterns from a single seed

cell, which is an asymmetric configuration, and update their cells at random; in the terms of Corollary 8.2, they restrict the inputs and drop determinism and synchrony.

- *Materials science.* Enumerations of the derivative superstructures of an alloy discard the "super-periodic" labelings, whose period is shorter than the supercell [35]. For a Bravais parent lattice, σ atomic species, and the supercell spanned by n times the parent basis vectors, these are exactly the configurations of S_{n^d} ($d = 3$ for bulk crystals), so Theorem 4.4 counts them, and for binary alloys Theorem 5.1 counts them at fixed composition. A general supercell has a quotient of the form $\mathbb{Z}_a \times \mathbb{Z}_b \times \mathbb{Z}_c$ (Smith normal form), a rectangular torus, whose super-periodic labelings Theorem 4.7 counts (for binary alloys at fixed composition, Remark B.2(i)).

- *Physics.* In exact diagonalization of quantum lattice models with translation symmetry, basis states whose period is shorter than the lattice need a modified normalization and are compatible with only some momenta [36, §§4.1.3, 4.4.1]; on $\mathbb{Z}_n^d$ with σ local states, these are exactly the configurations of S_{n^d}. More generally, by the argument of Theorem 8.1, any deterministic update map on a periodic grid that commutes with the grid translations (for example, a synchronous update applying the same rule at every site, as in a lattice gas or an explicit discretization of a reaction–diffusion equation) preserves, in exact arithmetic, every translation symmetry of its initial data; symmetric initial data must therefore be perturbed (round-off can act as such a perturbation) for symmetry breaking to be observed in such simulations.

- *Random initial states.* In cellular-automaton pseudorandom generators [37] and in simulations started from random configurations, a seed in S_{n^d} confines the whole trajectory to S_{n^d} (Theorem 8.1). For a uniform seed this happens with probability at most $(2^d - 1 + 2\log_2 n)\,\sigma^{-n^d/2}$ (Corollary 7.1), which is negligible except on very small lattices. Structured or sparse seeds are another matter (the all-zero seed lies in S_{n^d}). For a uniform seed with a prescribed number k of active cells, Theorem 5.1 gives the exact probability $|S^a_{n^d,k}|/\big(\binom{N}{k}(\sigma-1)^{N-k}\big)$, and Corollary 7.2 gives it when k is itself uniform; the algorithm of Section 9 tests a seed in $O(\mathrm{sopf}_d(n) \cdot n^d)$ cell reads.

Verification and its audit. A formal proof certifies a paper only as far as its formal statements say what the paper says: the kernel checks the proofs, not the reading. Besides the fidelity reviews of Section 11, the correspondence between every theorem, lemma, proposition, and corollary of this paper and its frozen Lean statements was therefore audited by hand by one of us (T.L.), with a viewer built for the purpose.[4] The viewer shows the official PDF beside the Lean declarations and links sub-expressions of the paper to the Lean sub-expressions claimed to express them (529 links for the 27 items), each link classified by kind: identical, a change of representation, a definition, a hypothesis, or a generalization (a few links point to a citation or have a counterpart on one side only). No link is of the kind "weaker", which would mark a part of the paper not covered by a frozen statement. The highlighted regions are computed from an instrumented copy of the paper whose words must typeset at exactly the positions they have in the official PDF; every action of the auditor is recorded, with timestamps, in a log kept under version control; and when the viewer data are rebuilt, a verdict is flagged as stale if the item's TEX source, the Lean source of its declarations, or its link data differ from those it was given against. All 27 items were approved. A verdict given on an earlier draft was carried over by a script only where the item's TEX source, the Lean source of its declarations, and its links were unchanged apart from line numbers; otherwise the item was judged again. A separate check of the link data, made during the audit, found that one clause, the symmetry-breaking sentence of Corollary 8.2, which the links had classified as having no Lean counterpart, was not covered by any frozen statement; it was reworded and frozen, and its new links were then audited. Tools

[4] The viewer and the audit log are in `research/formal/correspondence_audit/` of the project repository; T.L. is developing a product version of the viewer (`https://functorial.inc/`).

of this kind make the alignment between informal and formal statements inspectable item by item, and we regard them as a natural companion to proof assistants for papers like this one.

Open problems. (1) A theorem (not only computation) delimiting for which (n, d) the prime-dip/even-peak pattern of P^{unif} holds. (2) Rectangular tori $\mathbb{Z}_{n_1} \times \cdots \times \mathbb{Z}_{n_d}$, and finite abelian groups in general (counted by Theorem 4.7): a lower envelope for the number of shift-symmetric configurations that reduces to $F_d(n, \sigma)$ of Theorem 6.1 on $\mathbb{Z}_n^d$ and is attained exactly on a natural class of groups. (For a fixed order alone, the minimum is attained by the cyclic group, Corollary B.1.)

Appendices

A Proof of Theorem 6.1

Throughout, $q^m(c)$ denotes the truncation of (2) to the first m primes with exponents scaled by c:

$$q^m(c) \;=\; \sum_{\substack{\mathbf{v}\in\{0,\dots,d\}^m \\ \mathbf{v}\neq 0}} (-1)^{1+|\mathbf{v}|} \prod_{i=1}^{m} r_d(p_i, v_i)\, \sigma^{\,cN \prod_i p_i^{-v_i}}, \qquad q^m := q^m(1),$$

so $q^k = |S_{n^d}|$ by Theorem 4.4. We write $p := p_{m+1}$, $N_j := n^d/p^j$, $t := n/p$, $D := d(d-1)/2$.

If n is prime, Theorem 6.1 holds with equality by Corollary 4.5. For composite n we prove $q^1 \le q^2 \le \cdots \le q^k$ (Lemma A.3) and $q^1 > F_d(n, \sigma)$ (Lemma A.4 for $d \ge 3$; $d \le 2$ below), which give strict inequality at every composite n.

Lemma A.1 (Union interpretation on rectangular tori)**.** *Let $1 \le m < k$ and $1 \le j \le d$. Then*

$$0 \;\le\; q^m(p^{-j}) \;\le\; \sigma^{N_j}, \qquad \text{and} \qquad q^m(p^{-1}) \;\le\; \sum_{i\le m} \begin{bmatrix} d \\ 1 \end{bmatrix}_{p_i} \sigma^{N_1/p_i}.$$

Proof. Consider $T_j = (\mathbb{Z}/n\mathbb{Z})^{d-j} \times (\mathbb{Z}/\frac{n}{p}\mathbb{Z})^j$ with N_j cells. For $i \le m$, p_i divides both n and n/p (as $p_i \neq p$), so the p_i-torsion of T_j is again $\cong \mathbb{F}_{p_i}^d$ and Lemmas 3.1, 3.3–3.5, 4.1 and 4.2 apply verbatim on T_j. By the computation in the proof of Theorem 4.4, restricted to $p_1, \dots, p_m$, $q^m(p^{-j})$ equals the number of configurations of T_j fixed by some order-p_i cyclic subgroup, $i \le m$ — a union of fixed-point sets. Nonnegativity, the trivial bound σ^{N_j}, and Boole's inequality (at $j = 1$) follow. Exponent integrality: $v_{p_i}(N_j) = d\,\alpha_{p_i} \ge d \ge v_i$. □

Lemma A.2 (Coefficient estimates)**.** *Let $q \ge 2$ be real, with $\begin{bmatrix} d \\ j \end{bmatrix}_q$ and $r_d(q, j) = q^{\binom{j}{2}} \begin{bmatrix} d \\ j \end{bmatrix}_q$ defined by the product formula. Then $\begin{bmatrix} d \\ j \end{bmatrix}_q \le 4q^{j(d-j)}$, $\begin{bmatrix} d \\ 1 \end{bmatrix}_q \le 2q^{\,d-1}$, and $\sum_{j=2}^{d} r_d(q, j) \le 4(d-1)\, q^D$.*

Proof. $\begin{bmatrix} d \\ j \end{bmatrix}_q = q^{j(d-j)} \prod_{i=1}^{j} \frac{1-q^{-(d-j+i)}}{1-q^{-i}} \le q^{j(d-j)} \prod_{i\ge 1}(1-2^{-i})^{-1}$, and $\prod_{i\ge 1}(1-2^{-i}) \ge \frac{1}{2} \cdot \frac{3}{4} \cdot \frac{7}{8}\big(1 - \sum_{i\ge 4} 2^{-i}\big) = \frac{147}{512} > \frac{1}{4}$; also $\begin{bmatrix} d \\ 1 \end{bmatrix}_q \le q^d/(q-1) \le 2q^{d-1}$. Hence $r_d(q, j) \le 4q^{j(2d-j-1)/2}$; the exponent is a concave parabola in j with vertex $d - \frac{1}{2}$, so over $2 \le j \le d$ its maximum is D, attained at $j = d$ (and $j = d-1$ for $d \ge 3$); the tail sum is at most $4(d-1)q^D$. □

Lemma A.3 (Monotone chain)**.** *For composite n with $k \ge 2$, $d \ge 1$, $\sigma \ge 2$, and $1 \le m < k$: $q^m \le q^{m+1}$.*

Proof. Splitting q^{m+1} by the value j of v_{m+1} and isolating, for $j \ge 1$, the term with $(v_1, \dots, v_m) = \mathbf{0}$:

$$q^{m+1} - q^m \;=\; \sum_{j=1}^{d} (-1)^{j+1} r_d(p, j)\, B_j, \qquad B_j := \sigma^{N_j} - q^m(p^{-j}) \;\in\; [0, \sigma^{N_j}] \tag{3}$$

by Lemma A.1. For $d = 1$ the sum is the single term $B_1 \geq 0$; assume $d \geq 2$. Since $m \geq 1$: $p = p_{m+1} \geq 3$, and $p \prod_{i \leq m} p_i \mid n$ gives $p \leq n/2$ and $t \geq 2$. We use:

(I) $\sigma^{N_1/2} \geq 4\, t^{\,d-1}$;

(II) $\sigma^{N_1 - N_2} \geq 8(d-1)\, p^D$.

Given (I): by Lemma A.1 and $\sum_{i \leq m} \begin{bmatrix} d \\ 1 \end{bmatrix}_{p_i} \leq \sum_i 2p_i^{d-1} \leq 2\big(\sum_i p_i\big)^{d-1} \leq 2\big(\prod_i p_i\big)^{d-1} \leq 2t^{\,d-1}$ (superadditivity of x^{d-1}; $\sum p_i \leq \prod p_i$ for distinct primes; $\prod_{i \leq m} p_i \leq n/p = t$),

$$B_1 \;\geq\; \sigma^{N_1} - 2t^{\,d-1}\sigma^{N_1/2} \;\geq\; \tfrac{1}{2}\sigma^{N_1}.$$

Given (II): dropping the positive $j \geq 3$ odd terms of (3), bounding $B_j \leq \sigma^{N_2}$ for $j \geq 2$, and using Lemma A.2 with $r_d(p, 1) \geq 1$: $q^{m+1} - q^m \geq \frac{1}{2}\sigma^{N_1} - 4(d-1)p^D\sigma^{N_2} \geq 0$.

It remains to prove (I), (II) for composite $n \geq 6$, $d \geq 2$. (I): $N_1/2 = \frac{t}{2}n^{d-1} \geq n^{d-1}$ and $t \leq n/3$, so it suffices that $n^{d-1} \geq 2 + (d-1)\log_2(n/3)$; since $\log_2 n \leq n/2$ for $n \geq 4$, it suffices that $n^{d-1} \geq 2 + (d-1)n/2$ — for $d = 2$ this is $n \geq 4$; for $d \geq 3$, $n^{d-1} \geq 6^{d-2}n \geq (\frac{d-1}{2})n + 2$ since $6^{d-2} \geq \frac{d-1}{2} + 1$ (at $d = 3$: $6 \geq 2$; per increment of d the left side sextuples while the right grows by $\frac{1}{2}$). (II): $N_1 - N_2 = N_1(1 - \frac{1}{p}) \geq \frac{2}{3}N_1 \geq \frac{4}{3}n^{d-1}$ and $p^D \leq (n/2)^D$, so it suffices that $\frac{4}{3}n^{d-1} \geq 3 + \log_2(d-1) + D\log_2(n/2)$. For $d = 2$: $\frac{4}{3}n \geq 3 + \log_2(n/2)$ holds for $n \geq 6$. For $d \geq 3$, using $\log_2(d-1) \leq d-1$ and $\log_2(n/2) \leq n/2$, it suffices that $\frac{4}{3}\,6^{d-2}n \geq 2 + d + \frac{d(d-1)}{4}n$, which holds for $n \geq 6$ because $\frac{4}{3}\,6^{d-2} \geq \frac{d(d-1)}{4} + \frac{2+d}{6}$: at $d = 3$ this reads $8 \geq \frac{3}{2} + \frac{5}{6}$, and the induction step $6\big(\frac{d(d-1)}{4} + \frac{2+d}{6}\big) \geq \frac{(d+1)d}{4} + \frac{3+d}{6}$ reduces to $5d \geq 7$. □

Lemma A.4 (Base case, $d \geq 3$)**.** *For composite n, $\sigma \geq 2$, $d \geq 3$: $q^1 > F_d(n, \sigma)$.*

Proof. Let p_1 be the smallest prime factor; $p_1 \leq n/2$ and $p_1^2 \leq n$ (for $n = p_1^\alpha$, $\alpha \geq 2$; otherwise $p_1^2 < p_1 q \leq n$ for the second prime q). Dropping positive $r \geq 3$ terms of q^1 and negative terms of F_d, and using Lemma A.2 at $q = p_1$ and at the composite argument $q = n$ together with $\sigma^{n^d/p_1^r} \leq \sigma^{n^d/p_1^2}$ ($r \geq 2$) and $\sigma^{n^{d-r}} \leq \sigma^{n^{d-3}}$ ($r \geq 3$ odd):

$$q^1 \geq \begin{bmatrix} d \\ 1 \end{bmatrix}_{p_1} \sigma^{n^d/p_1} - 4(d-1)p_1^D\sigma^{n^d/p_1^2}, \qquad F_d \leq \begin{bmatrix} d \\ 1 \end{bmatrix}_n \sigma^{n^{d-1}} + 4(d-1)n^D\sigma^{n^{d-3}}.$$

All three exponent gaps are at least n^{d-1}: $\frac{n^d}{p_1} - \frac{n^d}{p_1^2} \geq \frac{n^d}{2p_1} \geq n^{d-1}$ and $\frac{n^d}{p_1} - n^{d-1} \geq n^{d-1}$ (as $n/p_1 \geq 2$), a fortiori for n^{d-3}. With $\begin{bmatrix} d \\ 1 \end{bmatrix}_{p_1} \geq 3$, $\begin{bmatrix} d \\ 1 \end{bmatrix}_n \leq 2n^{d-1}$, $p_1 \leq n$:

$$q^1 - F_d \;\geq\; \sigma^{n^d/p_1}\Big[3 - \big(8(d-1)n^D + 2n^{d-1}\big)2^{-n^{d-1}}\Big] \;\geq\; \sigma^{n^d/p_1} > 0,$$

provided $2^{n^{d-1}} \geq 12d\, n^D$ (using $8(d-1)n^D + 2n^{d-1} \leq 12dn^D$, as $n^{d-1} \leq n^D$ for $d \geq 2$). Taking logarithms it suffices that $n^{d-1} \geq 4 + d + \frac{d(d-1)}{2}\log_2 n$ for $n \geq 4$, $d \geq 3$: at $(4, 3)$ it reads $16 \geq 13$; for fixed d the margin increases in n (derivative $(d-1)n^{d-2} - \frac{d(d-1)}{2n\ln 2} > 0$), and the step $d \to d+1$ adds $1 + d\log_2 n \leq 1 + n^{d-1} \leq (n-1)n^{d-1}$ to the right side while multiplying the left by n. □

Proof of Theorem 6.1. $d = 1$: $F_1 = \sigma$; primes give $|S_n| = \sigma$, composites $|S_n| \geq \sigma^{n/p_1} \geq \sigma^2 > \sigma$. $d = 2$: $F_2(n, \sigma) = (n+1)\sigma^n - n\sigma$ and the claim is [1, Lemma E.21]; we import its base step ($q^1 > (n+1)\sigma^n - n\sigma$ at composite n, final display of the source proof; the inequality holds at every composite n, including the prime powers p^α, $\alpha \geq 2$, which the case split of the source ("If $k = 1$, i.e., n is a prime") does not treat (corrected in the companion erratum), and it is machine-checked, Section 11), while the chain step is re-proved uniformly by Lemma A.3 with $d = 2$ (the published chain, Lemma A.21 of [1], has three exceptional lattice sizes, repaired in the companion erratum). $d \geq 3$: prime n gives equality (Corollary 4.5); composite n with $k = 1$ gives $|S| = q^1 > F_d$ (Lemma A.4); composite n with $k \geq 2$ gives $|S| = q^k \geq \cdots \geq q^1 > F_d$ (Lemmas A.3, A.4). □

Every inequality in this appendix was verified exactly on parameter grids before submission (decomposition identity, torus sandwich, union bound, coefficient estimates, conditions (I)/(II), the chain, and the base case; see the reproducibility paragraph of Section 10), and the argument passed independent adversarial review.

B Finite abelian groups: dependence on the ranks

Notation is as in Section 4.5.

Corollary B.1 (Dependence on the ranks)**.** *Let $\sigma \geq 2$, and write $\mathrm{ac}(G) = \sigma^{|G|} - |S_G|$ for the number of aperiodic configurations.*

(i) $|S_G|$ depends on G only through N and its p-ranks; for example, $\mathbb{Z}_2 \times \mathbb{Z}_{24}$ and $\mathbb{Z}_4 \times \mathbb{Z}_{12}$ both have $50\,388\,256$ shift-symmetric configurations for $\sigma = 2$.

(ii) If G' has the same order as G, $d_p(G') = d_p(G) + 1$ for one prime p, and the other ranks agree, then $|S_{G'}| - |S_G| = p^{d_p(G)}\,\mathrm{ac}(G'') > 0$ for any abelian group G'' of order N/p with the ranks of G. In particular, among the abelian groups of order N, the cyclic group has the fewest shift-symmetric configurations.

Proof. (i) is immediate from Theorem 4.7. (ii) In complement form,

$$\mathrm{ac}(G) = \sum_{\mathbf{0} \leq \mathbf{v} \leq (d_p)_p} \prod_p c(p, d_p, v_p)\, \sigma^{N/\prod_p p^{v_p}}, \qquad c(p, e, v) = (-1)^v p^{\binom{v}{2}} \begin{bmatrix} e \\ v \end{bmatrix}_p .$$

The q-Pascal rule $\begin{bmatrix} e+1 \\ v \end{bmatrix}_p = \begin{bmatrix} e \\ v \end{bmatrix}_p + p^{e+1-v} \begin{bmatrix} e \\ v-1 \end{bmatrix}_p$ gives

$$c(p, e, v) - c(p, e+1, v) = p^e\, c(p, e, v-1), \qquad c(p, e, e+1) = 0.$$

Subtracting the two sums term by term and shifting v_p by one gives

$$\mathrm{ac}(G) - \mathrm{ac}(G') = p^{d_p(G)}\,\mathrm{ac}(G'').$$

Finally, $\mathrm{ac}(G'') \geq 1$: the indicator of a single cell has trivial stabilizer when $|G''| \geq 2$, and $\mathrm{ac}(G'') = \sigma$ when $|G''| = 1$. Write α_p for the exponent of p in N. An abelian group of order N has $1 \leq d_p \leq \alpha_p$ for every $p \mid N$, and for $1 \leq r \leq \alpha$ the group $\mathbb{Z}_{p^{\alpha-r+1}} \times \mathbb{Z}_p^{r-1}$ has order p^α and p-rank r; so G'' exists (as $d_p(G) + 1 \leq \alpha_p$), every rank profile with $1 \leq d_p \leq \alpha_p$ occurs, and every abelian group of order N is reached from the cyclic group, which has $d_p = 1$ for every $p \mid N$, by such steps. □

Remark B.2. (i) The k-active counts of Theorem 5.1 extend to G in the same way, with r_d replaced by r_{d_p} and n^d by N. (ii) G acts freely on the aperiodic configurations, so $N \mid \sigma^N - |S_G|$, and $(\sigma^N - |S_G|)/N$ is the number of G-orbits of aperiodic configurations; for rectangular tori these are the atranslational necklaces of [8].

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